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Research paper

Dynamics of a periodic Chikungunya model with temperature and rainfall effects☆

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ABSTRACT

In this paper, a periodic Chikungunya model with temperature and rainfall effects is proposed and studied, which incorporates time-dependent extrinsic incubation period, time-dependent maturation delay, asymptomatic and symptomatic infectious humans. Two threshold parameters for the extinction and persistence of mosquitos and the virus are derived, respectively: the mosquito reproduction number $\mathcal{R}_m$ and the basic reproduction number $\mathcal{R}_0$. Then the analytic results are verified by numerical simulation with the temperature and rainfall data of the state of Ceará, where the largest outbreak in Brazil's history occurred in 2017. And the effects of rainfall, seasonality, and asymptomatic infection in humans on mosquito population and the Chikungunya transmission are explored. Our simulations show that if these factors are not taken into account, the number of mosquito population and people infected may be overestimated. Finally, the relationships between $\mathcal{R}_m$ and $\mathcal{R}_0$ and some parameters are established.

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1. Introduction

Chikungunya virus (CHIKV) is a persistent and recurrent mosquito-borne alphavirus in tropical and subtropical regions, which badly endangers the health of people of different ages [1]. The intrinsic incubation period in humans is 1 to 12 days after the effective bite of an infected mosquito [2]. About 72%–97% of infected people are symptomatic, and more infected people need medical care than most people infected with arboviruses [3,4]. The main symptoms of the typical acute phase are high fever and polyarthralgia, however, other clinical symptoms may occur in the form of atypical infection, which makes the diagnosis of CHIKV difficult [4]. So far, there is no effective vaccine to prevent CHIKV infection and it is usually treated for symptomatic CHIKV patients [5,6]. While the diseased death rate is low, joint pain can last for months to years, seriously affecting their lives [2].

In the last ten years, CHIKV widely spread in countries in Africa, Asia, Europe, the Indian and Pacific Oceans, and the Americas in particular has seen unexpected outbreaks of the disease [4,7]. In December 2013, CHIKV broke out in St. Martin

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Island and then spread to the whole Americas [8]. Since then, forty-five countries have reported cases of autochthonous transmission [9], and 1.3 billion people in the world are at risk of contracting the disease [10]. In 2015, Centers for Disease Control and Prevention declared that Chikungunya disease was a notifiable disease [11].

The global warming, lack of herd immunity, poor living conditions and increased mobility of people in the areas of the outbreak have all contributed to the rapid spread of the disease in the Americas [12]. Furthermore, the increase in temperature and rainfall associated with global warming is directly related to the large number of Aedes aegypti, the main vector of CHIKV [13]. For instance, rising temperatures shorten the extrinsic incubation period in mosquitoes, leading to the increased mosquito reproduction, speeding up the mosquito oviposition cycle, and changing the mosquito feeding behavior [14,15]. Unlike other mosquito species, Aedes aegypti mosquitoes usually lay their eggs above the surface of the water, these eggs hatch only when the water level rises and wets them, and rainfall triggers the hatching of these eggs [16]. Therefore, it is necessary and practical to study how changes in temperature and rainfall affect the dynamics of Aedes aegypti mosquitoes. For the study of complexity in rainfall phenomena, we refer to [17].

The transmission of CHIKV between humans and mosquitoes has been mathematically modeled and analyzed over the past decade. In 2008, Dumont et al. established an ODE model of Chikungunya and verified it with the data from the 2005 outbreak on Réunion island [18]. Later, many scholars studied the transmission dynamics of CHIKV from different aspects (see [19–28] and the references cited therein). Considering climate change, Ruiz-Moreno et al. [29] proposed a climate-based stochastic model of mosquito population dynamics to analyze the potential risk of CHIKV entering the United States. Furthermore, Liu and Techlinski [30] developed a Chikungunya model with time-dependent contact rate and mosquito birth rate to explore the optimal pulse control strategy. In 2019, Satya et al. [31] constructed a temperature-dependent $\mathcal{R}_0$ model to study the risk of CHIKV transmission in India. Since about 3%-28% of infected people are asymptomatic, which has a significant impact on the spread of the disease, in terms of whether or not the infectious person has symptoms, Yakob and Clements [24] formulated a deterministic model of CHIKV spread between humans and mosquitoes, and the model's prediction fits well with the Réunion epidemic. Besides, Agusto et al. [27,28] also divided infectious humans into symptomatic and asymptomatic compartments and developed ODE models of CHIKV with three different human age compartments, these models are more in line with the characteristics of disease transmission.

Note that few mathematical models of Chikungunya transmission take into account both temperature and rainfall. Therefore, we develop the aforementioned works in three major aspects: (1) incorporating the temperature and rainfall effects into the model of Chikungunya transmission; (2) considering the time-dependent maturation delay, time-dependent extrinsic incubation period and constant intrinsic incubation period; and (3) dividing the infected population into asymptomatic and symptomatic compartments, and considering treatment strategy for symptomatic individuals.

The rest of the paper is organized as follows. In Section 2, we present a periodic Chikungunya model with temperature and rainfall effects. In Section 3, we introduce the mosquito reproduction number $\mathcal{R}_m$ and the basic reproduction number $\mathcal{R}_0$, and obtain the threshold dynamics in terms of $\mathcal{R}_m$ and $\mathcal{R}_0$. In Section 4, we carry out numerical simulations for Chikungunya spread in the state of Ceará, Brazil. A brief discussion section concludes the paper.

2. Model formulation

We assume that the human population is divided into six sub-classes: susceptible humans S_h, exposed humans E_h, asymptomatic infectious humans A_h, symptomatic infectious humans I_h, hospitalized humans P_h, and recovered humans R_h. Ae. aegypti mosquitoes are also divided into six sub-classes: dry eggs E_d, wet eggs E_w, larvae L_m, susceptible mature female mosquitoes S_m, exposed mature female mosquitoes E_m, and infectious mature female mosquitoes I_m. Dry eggs are those that have not come into contact with water and therefore cannot hatch, while wet eggs are those that come in contact with water. According to Valdez et al. [16], we assume that the water only comes from rain, and only wet eggs develop into larvae.

Since humans are endotherms, we assume that all human-related coefficients are positive constants. Suppose the recruitment rate and natural death rate of humans are Λ_h and μ_h. We consider the standard infection rate between mosquitoes and humans. Let α_h and α_m be the transmission probabilities from infective mosquitos to humans and from infective humans to mosquitos, respectively. Since the bite rate $\beta(t)$ is greatly affected by climate change, we assume that this rate is periodic. Let $M_h(t)$ be the numbers of the exposed humans moving in the infected class at time t, a fraction ρ of which are the asymptomatic A_h, and the remaining ones are the symptomatic I_h. The symptomatic infectious humans are hospitalized at a rate η_h. Let q_1, q_2 and q_3 be the recover rates of the asymptomatic A_h, symptomatic I_h, hospitalized P_h, respectively.

To explore the temperature and rainfall effects, we assume that all mosquito-related coefficients are positive and ω-periodic functions with $\omega = 12$ months. Let $\mu_b(t)$, $K_d(t)$, $\kappa_d(t)$ and $\eta_w(t)$ be the birth rate of dry eggs, maximal dry egg capacity, fraction of dry eggs that become wet eggs and maturation rate of larvae, respectively. The mortality rates for eggs, larvae, and adult mosquitoes are represented by $\mu_E(t)$, $\mu_L(t)$ and $\mu_m(t)$, respectively. Let $M_L(t)$ and $M_m(t)$ be the numbers of newly mature mosquitoes and infectious mosquitoes at time t, respectively.

Based on the aforementioned assumptions, a schematic diagram of the Chikungunya transmission between humans and mosquitoes is given in Fig. 1, and the biological interpretation of model coefficients is shown in Table 1. Motivated by the observation that the total population $N_h(t)$ stabilizes at a steady state, we assume that $N_h(t) \equiv N_h^* := \Lambda_h/\mu_h$ for the sake of

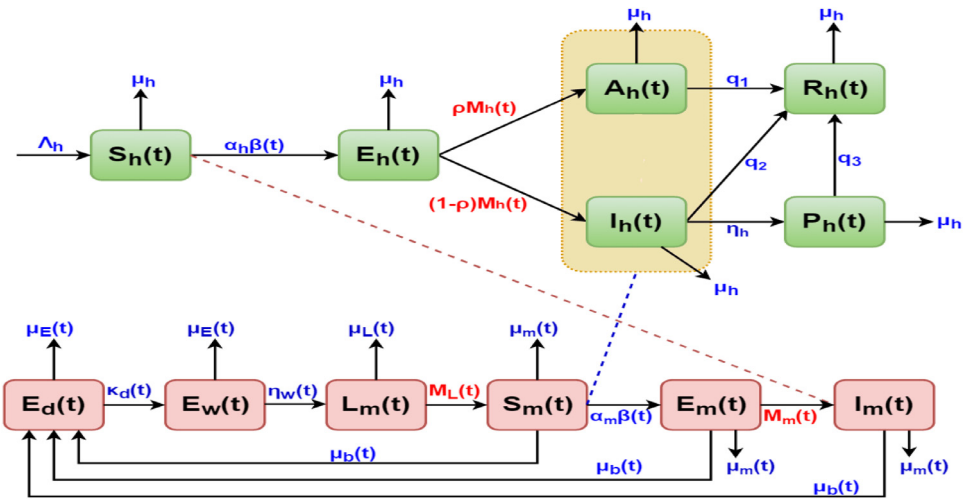


Fig. 1. Schematic diagram for the disease transmission.

Table 1
Biological explanations for parameters of system (2.1).

Parameters	Description
Λ_h	Recruitment rate of humans
$1/\mu_h$	Average lifespan of humans
ρ	Fraction of exposed humans becoming asymptomatic
$\beta(t)$	Bite rate of mosquitos
α_h	Transmission probability per bite from infective mosquitos to humans
α_m	Transmission probability per bite from infective humans to mosquitos
τ_h	The intrinsic incubation period (IIP) in humans
$\tau_m(t)$	The extrinsic incubation period (EIP) in mosqiutios
q_1	Recovery rate of asymptomatic infectious humans
q_2	Recovery rate of symptomatic infectious humans
q_3	Recovery rate of hospitalized humans
η_h	Hospitalizing rate of symptomatic infectious humans
$K_d(t)$	The maximal dry egg capacity
$\mu_b(t)$	Birth rate of egg
$\kappa_d(t)$	Fraction of dry eggs that become wet eggs
$\eta_w(t)$	Maturation rate of wet eggs
$\tau_w(t)$	Maturation delay of larvae
$\mu_E(t)$	Mortality rate of eggs
$\mu_L(t)$	Mortality rate of larvae
$\mu_m(t)$	Mortality rate of mature female mosquitos

simplicity. Therefore, we have the following system:

$$\begin{cases}
\frac{dS_h}{dt}(t) = \Lambda_h - \alpha_h\beta(t)\frac{I_m(t)}{N_h^*}S_h(t) - \mu_h S_h(t),\\
\frac{dE_h}{dt}(t) = \alpha_h\beta(t)\frac{I_m(t)}{N_h^*}S_h(t) - M_h(t) - \mu_h E_h(t),\\
\frac{dA_h}{dt}(t) = \rho M_h(t) - (q_1+\mu_h)A_h(t),\\
\frac{dI_h}{dt}(t) = (1-\rho)M_h(t) - (\eta_h+q_2+\mu_h)I_h(t),\\
\frac{dP_h}{dt}(t) = \eta_h I_h(t) - (q_3+\mu_h)P_h(t),\\
\frac{dR_h}{dt}(t) = q_1A_h(t) + q_2I_h(t) + q_3P_h(t) - \mu_h R_h(t),\\
\frac{dE_d}{dt}(t) = \mu_b(t)\left(1-\frac{E_d(t)}{K_d(t)}\right)(S_m(t)+E_m(t)+I_m(t)) - (\kappa_d(t)+\mu_E(t))E_d(t),\\
\frac{dE_w}{dt}(t) = \kappa_d(t)E_d(t) - (\eta_w(t)+\mu_E(t))E_w(t),\\
\frac{dL_m}{dt}(t) = \eta_w(t)E_w(t) - M_L(t) - \mu_L(t)L(t),\\
\frac{dS_m}{dt}(t) = M_L(t) - \alpha_m\beta(t)\frac{A_h(t)+I_h(t)}{N_h^*}S_m(t) - \mu_m(t)S_m(t),\\
\frac{dE_m}{dt}(t) = \alpha_m\beta(t)\frac{A_h(t)+I_h(t)}{N_h^*}S_m(t) - M_m(t) - \mu_m(t)E_m(t),\\
\frac{dI_m}{dt}(t) = M_m(t) - \mu_m(t)I_m(t).
\end{cases} \tag{2.1}$$

Let τ_h be the intrinsic incubation period (IIP) of humans, $\tau_m(t)$ be the extrinsic incubation period (EIP) of mosquitoes, and $\tau_w(t)$ be the maturation time of the larvae. By the arguments similar to those in [32–34], we then obtain

$$\begin{aligned}
M_h(t) &= \alpha_h\beta(t-\tau_h)\frac{I_m(t-\tau_h)}{N_h^*}S_h(t-\tau_h)e^{-\mu_h\tau_h},\\
M_m(t) &= \alpha_m(1-\tau_m'(t))\beta(t-\tau_m(t))\frac{A_h(t-\tau_m(t))+I_h(t-\tau_m(t))}{N_h^*}\\
&\quad\times S_m(t-\tau_m(t))e^{-\int_{t-\tau_m(t)}^{t}\mu_m(\xi)d\xi},\\
M_L(t) &= (1-\tau_w'(t))\eta_w(t-\tau_w(t))E_w(t-\tau_w(t))e^{-\int_{t-\tau_w(t)}^{t}\mu_L(\xi)d\xi},
\end{aligned}$$

where the inequalities $1-\tau_m'(t)>0$ and $1-\tau_w'(t)>0$ hold true.

For simplicity, let

$$\beta_h(t)=\alpha_h\frac{\beta(t)}{N_h^*}\quad\text{and}\quad\beta_m(t)=\alpha_m\frac{\beta(t)}{N_h^*}.$$

Then

$$\beta_h(t-\tau_h)=\alpha_h\frac{\beta(t-\tau_h)}{N_h^*}\quad\text{and}\quad\beta_m(t-\tau_m(t))=\alpha_m\frac{\beta(t-\tau_m(t))}{N_h^*}.$$

Thus, system (2.1) is rewritten to the following one:

$$\begin{cases}
\frac{dS_h}{dt}(t)=\Lambda_h-\beta_h(t)S_h(t)I_m(t)-\mu_hS_h(t),\\
\frac{dE_h}{dt}(t)=\beta_h(t)S_h(t)I_m(t)-\beta_h(t-\tau_h)S_h(t-\tau_h)I_m(t-\tau_h)e^{-\mu_h\tau_h}-\mu_hE_h(t),\\
\frac{dA_h}{dt}(t)=\rho\beta_h(t-\tau_h)S_h(t-\tau_h)I_m(t-\tau_h)e^{-\mu_h\tau_h}-(q_1+\mu_h)A_h(t),\\
\frac{dI_h}{dt}(t)=(1-\rho)\beta_h(t-\tau_h)S_h(t-\tau_h)I_m(t-\tau_h)e^{-\mu_h\tau_h}-(\eta_h+q_2+\mu_h)I_h(t),\\
\frac{dP_h}{dt}(t)=\eta_hI_h(t)-(q_3+\mu_h)P_h(t),\\
\frac{dR_h}{dt}(t)=q_1A_h(t)+q_2I_h(t)+q_3P_h(t)-\mu_hR_h(t),\\
\frac{dE_d}{dt}(t)=\mu_b(t)\big(1-\frac{E_d(t)}{K_d(t)}\big)(S_m(t)+E_m(t)+I_m(t))-(\kappa_d(t)+\mu_E(t))E_d(t),\\
\frac{dE_w}{dt}(t)=\kappa_d(t)E_d(t)-(\eta_w(t)+\mu_E(t))E_w(t),\\
\frac{dL_m}{dt}(t)=\eta_w(t)E_w(t)-(1-\tau_w'(t))\eta_w(t-\tau_w(t))E_w(t-\tau_w(t))e^{-\int_{t-\tau_w(t)}^{t}\mu_L(\xi)d\xi}\\
\quad-\mu_L(t)L(t),\\
\frac{dS_m}{dt}(t)=(1-\tau_w'(t))\eta_w(t-\tau_w(t))E_w(t-\tau_w(t))e^{-\int_{t-\tau_w(t)}^{t}\mu_L(\xi)d\xi}-\mu_m(t)S_m(t)\\
\quad-\beta_m(t)(A_h(t)+I_h(t))S_m(t),\\
\frac{dE_m}{dt}(t)=\beta_m(t)(A_h(t)+I_h(t))S_m(t)-\mu_m(t)E_m(t)-(1-\tau_m'(t))\beta_m(t-\tau_m(t))\\
\quad\times(A_h(t-\tau_m(t))+I_h(t-\tau_m(t)))S_m(t-\tau_m(t))e^{-\int_{t-\tau_m(t)}^{t}\mu_m(\xi)d\xi},\\
\frac{dI_m}{dt}(t)=(1-\tau_m'(t))\beta_m(t-\tau_m(t))(A_h(t-\tau_m(t))+I_h(t-\tau_m(t)))\\
\quad\times S_m(t-\tau_m(t))e^{-\int_{t-\tau_m(t)}^{t}\mu_m(\xi)d\xi}-\mu_m(t)I_m(t).
\end{cases}\tag{2.2}$$

The biological compatibility conditions for system (2.2) are as follows:

$$\begin{aligned}
E_h(0) &= \int_{-\tau_h}^{0}\beta_h(s)I_m(s)S_h(s)e^{-\mu_hs}ds,\\
L_m(0) &= \int_{-\tau_w(0)}^{0}\eta_w(s)E_w(s)e^{-\int_s^0\mu_E(\xi)d\xi}ds,\\
E_m(0) &= \int_{-\tau_m(0)}^{0}\beta_m(s)(A_h(s)+I_h(s))S_m(s)e^{-\int_s^0\mu_m(\xi)d\xi}ds.
\end{aligned}\tag{2.3}$$

Similar to [35], we impose the following condition on $K_d(t)$, which is always assumed throughout this paper:

$$\frac{dK_d(t)}{dt}\geq-(\kappa_d(t)+\mu_E(t))K_d(t),\ \ \forall t\in\mathbb{R}.\tag{2.4}$$

When $K_d(t)$ is a positive constant, (2.4) holds automatically.

Since $E_h(t)$, $P_h(t)$, $R_h(t)$ and $L(t)$ do not appear in the other equations of system (2.2), it suffices to study the reduced system:

$$\begin{cases} \frac{dS_h}{dt}(t) = \Lambda_h - \beta_h(t)S_h(t)I_m(t) - \mu_h S_h(t), \\ \frac{dA_h}{dt}(t) = \rho\beta_h(t-\tau_h)S_h(t-\tau_h)I_m(t-\tau_h)e^{-\mu_h\tau_h} - (q_1+\mu_h)A_h(t), \\ \frac{dI_h}{dt}(t) = (1-\rho)\beta_h(t-\tau_h)S_h(t-\tau_h)I_m(t-\tau_h)e^{-\mu_h\tau_h} - (\eta_h+q_2+\mu_h)I_h(t), \\ \frac{dE_d}{dt}(t) = \mu_b(t)\left(1-\frac{E_d(t)}{K_d(t)}\right)(S_m(t)+E_m(t)+I_m(t)) - (\kappa_d(t)+\mu_E(t))E_d(t), \\ \frac{dE_w}{dt}(t) = \kappa_d(t)E_d(t) - (\eta_w(t)+\mu_E(t))E_w(t), \\ \frac{dS_m}{dt}(t) = (1-\tau_w'(t))\eta_w(t-\tau_w(t))E_w(t-\tau_w(t))e^{-\int_{t-\tau_w(t)}^{t}\mu_L(\xi)d\xi} \\ \quad -\beta_m(t)(A_h(t)+I_h(t))S_m(t) - \mu_m(t)S_m(t), \\ \frac{dE_m}{dt}(t) = \beta_m(t)(A_h(t)+I_h(t))S_m(t) - \mu_m(t)E_m(t) - (1-\tau_m'(t))\beta_m(t-\tau_m(t)) \\ \quad \times(A_h(t-\tau_m(t))+I_h(t-\tau_m(t)))S_m(t-\tau_m(t))e^{-\int_{t-\tau_m(t)}^{t}\mu_m(\xi)d\xi}, \\ \frac{dI_m}{dt}(t) = (1-\tau_m'(t))\beta_m(t-\tau_m(t))(A_h(t-\tau_m(t))+I_h(t-\tau_m(t))) \\ \quad \times S_m(t-\tau_m(t))e^{-\int_{t-\tau_m(t)}^{t}\mu_m(\xi)d\xi} - \mu_m(t)I_m(t). \end{cases} \tag{2.5}$$

3. Threshold dynamics

In this section, according to the theory developed in [36], we introduce the mosquito reproduction number $\mathcal{R}_m$ and the basic reproduction number $\mathcal{R}_0$, and then study the global dynamics of system (2.5).

3.1. Threshold dynamics in terms of $\mathcal{R}_m$

If Chikungunya disease does not exist, then the subsystem for mosquito population is as follows:

$$\begin{cases} \frac{dE_d}{dt}(t) = \mu_b(t)\left(1-\frac{E_d(t)}{K_d(t)}\right)S_m(t) - (\kappa_d(t)+\mu_E(t))E_d(t), \\ \frac{dE_w}{dt}(t) = \kappa_d(t)E_d(t) - (\eta_w(t)+\mu_E(t))E_w(t), \\ \frac{dS_m}{dt}(t) = (1-\tau_w'(t))\eta_w(t-\tau_w(t))E_w(t-\tau_w(t))e^{-\int_{t-\tau_w(t)}^{t}\mu_L(\xi)d\xi} - \mu_m(t)S_m(t). \end{cases} \tag{3.1}$$

Linearizing system (3.1) at (0,0,0), we obtain a cooperative periodic system:

$$\begin{cases} \frac{dE_d}{dt}(t) = \mu_b(t)S_m(t) - (\kappa_d(t)+\mu_E(t))E_d(t), \\ \frac{dE_w}{dt}(t) = \kappa_d(t)E_d(t) - (\eta_w(t)+\mu_E(t))E_w(t), \\ \frac{dS_m}{dt}(t) = (1-\tau_w'(t))\eta_w(t-\tau_w(t))E_w(t-\tau_w(t))e^{-\int_{t-\tau_w(t)}^{t}\mu_L(\xi)d\xi} - \mu_m(t)S_m(t). \end{cases} \tag{3.2}$$

Let $\tilde{\tau}_1 = \max_{0\le t\le\omega}\tau_w(t)$ and $C = C([-\tilde{\tau}_1, 0], \mathbb{R}^3)$ be a Banach space with the norm $\|\phi\| = \max_{\theta\in[-\tilde{\tau}_1,0]}\|\phi(\theta)\|_{\mathbb{R}^3}$. Then (C, C^+) is an ordered Banach space, where $C^+ = C([-\tilde{\tau}_1, 0], \mathbb{R}^3_+)$.

For any $t\in\mathbb{R}$, $(\varphi_1, \varphi_2, \varphi_3)\in C$, we define the mapping $\tilde{F}:\mathbb{R}\to\mathcal{L}(C,\mathbb{R}^3)$ by

$$\tilde{F}(t)\begin{pmatrix}\varphi_1\\ \varphi_2\\ \varphi_3\end{pmatrix} = \begin{pmatrix} \mu_b(t)\varphi_3(0) \\ \kappa_d(t)\varphi_1(0) \\ (1-\tau_w'(t))\eta_w(t-\tau_w(t))e^{-\int_{t-\tau_w(t)}^{t}\mu_L(\xi)d\xi}\varphi_2(-\tau_w(t)) \end{pmatrix}.$$

Let

$$\tilde{V}(t) = \begin{pmatrix} \kappa_d(t)+\mu_E(t) & 0 & 0 \\ 0 & \eta_w(t)+\mu_E(t) & 0 \\ 0 & 0 & \mu_m(t) \end{pmatrix},$$

and $\tilde{Z}(t,k), t\ge k$, be the evolution matrix of the linear periodic system $\frac{d\tilde{W}(t)}{dt} = -\tilde{V}(t)\tilde{W}(t)$, that is,

$$\frac{\partial}{\partial t}\tilde{Z}(t,k) = -\tilde{V}(t)\tilde{Z}(t,k), \quad \forall t\ge k, \quad \text{and} \quad \tilde{Z}(k,k) = I, \quad \forall k\in\mathbb{R},$$

where I is the 3×3 identity matrix. It can easily be seen that

$$\tilde{Z}(t,k) = \begin{pmatrix} e^{-\int_k^t(\kappa_d(\xi)+\mu_E(\xi))d\xi} & 0 & 0 \\ 0 & e^{-\int_k^t(\eta_w(\xi)+\mu_E(\xi))d\xi} & 0 \\ 0 & 0 & e^{-\int_k^t\mu_m(\xi)d\xi} \end{pmatrix}.$$

Let C_ω be the ordered Banach space of all continuous and ω-periodic functions from $\mathbb{R}$ to $\mathbb{R}^3$, equipped with the maximum norm and the positive cone $C_\omega^+ := \{u\in C_\omega : u(t)\ge 0, \forall t\in\mathbb{R}\}$.

Following [36], we define the next generation operator $\tilde{L}$ on C_ω:

$$[\tilde{L}u](t)=\int_0^\infty \tilde{Z}(t,t-k)\tilde{F}(t-k)u(t-k+\cdot)dk,\quad \forall t\in\mathbb{R}, u\in C_\omega,$$

and the mosquito reproduction number

$$\mathcal{R}_m := r(\tilde{L}), \tag{3.3}$$

where $r(\tilde{L})$ is the spectral radius of $\tilde{L}$. Let $\tilde{\mathcal{P}}_m(t)$ be the solution map of system (3.2) and $r(\tilde{\mathcal{P}}_m)$ be the spectral radius of poincaré map $\tilde{\mathcal{P}}_m := \tilde{\mathcal{P}}_m(\omega)$ for system (3.2) on C. By [36, Theorem 2.1], we obtain the following lemma.

Lemma 3.1. *$\mathcal{R}_m - 1$ has the same sign as $r(\tilde{\mathcal{P}}_m) - 1$, where $\mathcal{R}_m$ is defined in (3.3).*

Let

$$\mathcal{A}(t)=[0,K_d(t)]\times C([-\tilde{\tau}_1,0],\mathbb{R}_+)\times\mathbb{R}_+,\ \forall t\geq 0.$$

Then we have the following result for system (3.1), which can be proved in a similar way to Lemma 3.6.

Lemma 3.2. *For any $\nu\in\mathcal{A}(0)$, system (3.1) has a unique nonnegative solution $x(t,\nu)$ with $x_0=\nu$, and $x_t(\nu):=(x_1(t,\nu),x_{2t}(\nu),x_3(t,\nu))\in\mathcal{A}(t)$ on $[0,+\infty)$.*

Let

$$\mathcal{B}(t)=[0,K_d(t)]\times C([-\tau_w(0),0],\mathbb{R}_+)\times\mathbb{R}_+,\ \forall t\geq 0.$$

Lemma 3.3. *For any $u\in\mathcal{B}(0)$, system (3.1) has a unique nonnegative solution $y(t,u)$ with $y_0=u$, and $y_t(u):=(y_1(t,u),y_{2t}(u),y_3(t,u))\in\mathcal{B}(t)$ on $[0,+\infty)$.*

Proof. Define $\bar{\tau}_1=\min_{0\leq t\leq\omega}\tau_w(t)$. For any $t\in[0,\bar{\tau}_1]$, since $t-\tau_w(t)$ is strictly increasing with respect to t, we have

$$-\tau_w(0)\leq t-\tau_w(t)\leq\bar{\tau}_1-\tau_w(\bar{\tau}_1)\leq\bar{\tau}_1-\bar{\tau}_1=0,$$

and hence, $y_2(t-\tau_w(t))=u_2(t-\tau_w(t))$. Thus, we get the following ODE system for $t\in[0,\bar{\tau}_1]$:

$$\begin{cases}\frac{dy_1}{dt}(t)=\mu_b(t)\big(1-\frac{y_1(t)}{K_d(t)}\big)y_3(t)-(\kappa_d(t)+\mu_E(t))y_1(t),\\ \frac{dy_2}{dt}(t)=\kappa_d(t)y_1(t)-(\eta_w(t)+\mu_E(t))y_2(t),\\ \frac{dy_3}{dt}(t)=(1-\tau_w'(t))\eta_w(t-\tau_w(t))u_2(t-\tau_w(t))e^{-\int_{t-\tau_w(t)}^t\mu_L(\xi)d\xi}-\mu_m(t)y_3(t).\end{cases}$$

Given $u\in\mathcal{B}(0)$, the above system admits the solution $(y_1(t),y_2(t),y_3(t))$ for $t\in[0,\bar{\tau}_1]$, that is, $y_2(\theta)=u_2(\theta)$ for $\theta\in[-\tau_w(0),\bar{\tau}_1]$ and $y_i(\theta)=u_i(\theta)$ for $\theta\in[0,\bar{\tau}_1]$, $i=1,3$.

By repeating this step to $t\in[n\bar{\tau}_1,(n+1)\bar{\tau}_1]$, $n=1,2,3,\cdots$, we then see that the solution $y(t,u)$ with $y_0=u$ of system (3.1) exists uniquely and $y_t(u):=(y_1(t,u),y_{2t}(u),y_3(t,u))\in\mathcal{B}(t)$ on $[0,+\infty)$. □

Remark 3.1. By the uniqueness of solutions in Lemmas 3.2 and 3.3, it follows that for any $\nu\in\mathcal{A}(0)$ and $u\in\mathcal{B}(0)$ with $\nu_1=u_1$, $\nu_3=u_3$, and $\nu_2(\theta)=u_2(\theta)$, $\forall\theta\in[-\tau_w(0),0)$, we have $x(t,\nu)=y(t,u)$, $t\geq 0$, where solutions $x(t,\nu)$ and $y(t,u)$ of system (3.1) satisfy $x_0=\nu$ and $y_0=u$, respectively.

Let $G_m(t)$ be the solution maps of system (3.1), that is, $G_m(t)u=y_t(u)$, $t\geq 0$, where the unique solution of system (3.1) $y(t,u)$ satisfies $y_0=u\in\mathcal{B}(0)$. As show in [37, Lemma 3.5], we see that $G_m(t):\mathcal{B}(0)\to\mathcal{B}(t)$ is an ω-periodic semiflow.

Let $r(P_m)$ be the spectral radius of poincaré map $P_m:=P_m(\omega)$ for system (3.2) on the space $\mathbb{R}_+\times C([-\tau_w(0),0],\mathbb{R}_+)\times\mathbb{R}_+$. Then we have the following threshold type results for system (3.1).

Lemma 3.4. *Let $\mathcal{R}_m$ be defined as in (3.3). Then the following two statements hold true:*

(i) If $\mathcal{R}_m\leq 1$, then (0,0,0) is globally attractive for system (3.1) in $\mathcal{B}(0)$;

(ii) If $\mathcal{R}_m>1$, then the unique positive ω-periodic solution $(E_d^*(t),E_w^*(t),S_m^*(t))$ of system (3.1) is globally attractive in $\mathcal{B}(0)\setminus\{(0,0,0)\}$.

Proof. For any given $\nu,\zeta\in\mathcal{B}(0)$ with $\nu>\zeta$, let $y(t,\nu)=(y_1(t),y_2(t),y_3(t))$ and $z(t,\zeta)=(z_1(t),z_2(t),z_3(t))$ be the unique solutions of system (3.1) with $y_0=\nu$ and $z_0=\zeta$, respectively. Without loss of generality, we assume that $\nu_2>\zeta_2$. Apparently, Lemma 3.3 implies that $G_m(t)$ is monotone for all $t\geq 0$. Next we prove that $G_m(t)$ is eventually strongly monotone.

Claim 1. There exists $t_0\in[0,\bar{\tau}_1]$ such that $y_3(t)>z_3(t)$, $\forall t\geq t_0$.

Claim 2. $y_1(t)>z_1(t)$, $\forall t\geq t_0$.

The above two claims can be proved by the arguments similar to Claims 1 and 3 in Lemma 3.8.

In view of Claims 1 and 2, it follows that $(y_1(t),y_2(t),y_3(t))\gg(z_1(t),z_2(t),z_3(t))$, $t\geq t_0$. Since $t_0\in[0,\bar{\tau}_1]$, it follows that $(y_1(t),y_{2t},y_3(t))\gg(z_1(t),z_{2t},z_3(t))$, $\forall t\geq t_0+\tau_w(0)$, that is, $y(t,\nu)\gg z(t,\zeta)$, $\forall t\geq t_0+\tau_w(0)$. This proves that $G_m(t)$ is strongly monotone for any $t\geq t_0+\tau_w(0)$.

For any given $v \gg 0$ in $\mathcal{B}(0)$ and $\lambda \in (0, 1)$, let $y(t, v)$ and $y(t, \lambda v)$ be the solutions of system (3.1) satisfying $y_0 = v$ and $y_0 = \lambda v$, respectively. Denote $\phi(t) = \lambda y(t, v)$ and $\varphi(t) = y(t, \lambda v)$. From Lemma 3.3, we have $\phi(t) > 0$ and $\varphi(t) > 0$ for all $t \geq 0$. Besides, $\phi_1(0) = \lambda v_1 = \varphi_1(0)$, $\phi_2(\theta) = \lambda v_2 = \varphi_2(\theta)$, $\phi_3(0) = \lambda v_3 = \varphi_3(0)$, $\forall \theta \in [-\tau_w(0), 0]$.

Since $-\tau_w(0) \leq t - \tau_w(0) \leq 0$, $\forall t \in [0, \bar{\tau}_1]$, it follows that $\phi_2(t - \tau_w(t)) = \lambda v_2(t - \tau_w(t)) = \varphi_2(t - \tau_w(t))$. Therefore, $\phi(t)$ satisfies the following differential inequality:

$$\begin{cases} \frac{d\phi_1}{dt}(t) < \mu_b(t)\left(1 - \frac{\phi_1(t)}{K_d(t)}\right)\phi_3(t) - (\kappa_d(t) + \mu_E(t))\phi_1(t), \\ \frac{d\phi_2}{dt}(t) < \kappa_d(t)\phi_1(t) - (\eta_w(t) + \mu_E(t))\phi_2(t), \\ \frac{d\phi_3}{dt}(t) < (1 - \tau_w'(t))\eta_w(t - \tau_w(t))\phi_2(t - \tau_w(t))e^{-\int_{t-\tau_w(t)}^{t}\mu_L(\xi)d\xi} - \mu_m(t)\phi_3(t), \end{cases}$$

for all $t \in [0, \bar{\tau}_1]$. By the comparison principle [38, Theorem 4] and $\phi(0) = \varphi(0)$, we obtain $\phi_i(t) < \varphi_i(t)$ for each $i = 1, 2, 3$ and $t \in [0, \bar{\tau}_1]$. Similarly, for any internal $(n\bar{\tau}_1, (n+1)\bar{\tau}_1)$, $n = 1, 2, 3, \cdots$, we have $\phi_i(t) < \varphi_i(t)$ for all $t > 0$ and each $i = 1, 2, 3$, namely, $y(t, \lambda v) \geq \lambda y(t, v)$, $\forall t > 0$. Thus, $y_t(\lambda v) \geq \lambda y_t(v)$, $\forall t > \tau_w(0)$.

Fix an integer $n_0 > \bar{\tau}_1 + \tau_w(0)$. Then $G_m^{n_0} = G_m(n_0\omega) : \mathcal{B}(0) \to \mathcal{B}(0)$ is strongly monotone and strictly subhomogeneous. Note that $DG_m^{n_0}(0) = DG_m(n_0\omega)(0) = P_m^{n_0}(\omega) = P_m^{n_0}$, and $r(P_m^{n_0}) = r(P_m)^{n_0}$. By applying [39, Theorem 2.3.4 and Lemma 2.2.1] to $G_m^{n_0}$ and combining [37, Lemma 3.8] and Lemma 3.1, we then have the following result:

(i) If $\mathcal{R}_m \leq 1$, then (0,0,0) is globally attractive for system (3.1) in $\mathcal{B}(0)$;

(ii) If $\mathcal{R}_m > 1$, then the unique positive $n_0\omega$-periodic solution $(E_d^*(t), E_w^*(t), S_m^*(t))$ of system (3.1) is globally attractive in $\mathcal{B}(0) \setminus \{(0, 0, 0)\}$.

Finally, we need to prove that $\tilde{y}(t) = y(t, v) = (E_d^*(t), E_w^*(t), S_m^*(t))$ is also an ω-periodic solution of system (3.1). Since $G_m^{n_0}(G(v)) = G(G_m^{n_0}(v)) = G(v)$, then the uniqueness of the positive fixed point of $G_m^{n_0}$ implies that $G(v) = v$. Therefore, $\tilde{y}(t)$ is an ω-periodic solution of system (3.1). □

3.2. Threshold dynamics in terms of $\mathcal{R}_0$

Letting $A_h = I_h = E_m = I_m = 0$ in system (2.5), we have the following system:

$$\begin{cases} \frac{dS_h}{dt}(t) = \Lambda_h - \mu_h S_h(t), \\ \frac{dE_d}{dt}(t) = \mu_b(t)\left(1 - \frac{E_d(t)}{K_d(t)}\right)S_m(t) - (\kappa_d(t) + \mu_E(t))E_d(t), \\ \frac{dE_w}{dt}(t) = \kappa_d(t)E_d(t) - (\eta_w(t) + \mu_E(t))E_w(t), \\ \frac{dS_m}{dt}(t) = (1 - \tau_w'(t))\eta_w(t - \tau_w(t))E_w(t - \tau_w(t))e^{-\int_{t-\tau_w(t)}^{t}\mu_L(\xi)d\xi} - \mu_m(t)S_m(t). \end{cases} \tag{3.4}$$

It is obvious that the first equation of system (3.4) has a globally stable positive equilibrium $S_h^* = \Lambda_h/\mu_h$ in $\mathbb{R}^+$. Then we can linearize system (2.5) at the disease-free periodic solution $(S_h^*, 0, 0, E_d^*(t), E_w^*(t), S_m^*(t), 0, 0)$ to obtain the following linear equations for the infective variables A_h, I_h and I_m:

$$\begin{cases} \frac{dA_h}{dt}(t) = -c_{11}(t)A_h(t) + c_{13}(t)I_m(t - \tau_h), \\ \frac{dI_h}{dt}(t) = -c_{22}(t)I_h(t) + c_{23}(t)I_m(t - \tau_h), \\ \frac{dI_m}{dt}(t) = c_{31}(t)(A_h(t - \tau_m(t)) + I_h(t - \tau_m(t))) - c_{33}(t)I_m(t), \end{cases} \tag{3.5}$$

where $c_{11}(t) = q_1 + \mu_h$, $c_{13}(t) = \rho\beta_h(t - \tau_h)S_h^*e^{-\mu_h\tau_h}$, $c_{22}(t) = \eta_h + q_2 + \mu_h$, $c_{23}(t) = (1 - \rho)\beta_h(t - \tau_h)S_h^*e^{-\mu_h\tau_h}$, $c_{33}(t) = \mu_m(t)$, and

$$c_{31}(t) = (1 - \tau_m'(t))\beta_m(t - \tau_m(t))S_m^*(t - \tau_m(t))e^{-\int_{t-\tau_m(t)}^{t}\mu_m(\xi)d\xi}.$$

Let $\tilde{\tau}_2 = \max\{\tau_h, \max_{0\leq t\leq\omega}\tau_m(t)\}$ and $C = C([-\tilde{\tau}_2, 0], \mathbb{R}^3)$ be a Banach space with the norm $\|\phi\| = \max_{\theta\in[-\tilde{\tau}_2,0]}\|\phi(\theta)\|_{\mathbb{R}^3}$. Then (C, C^+) is an ordered Banach space, where $C^+ = C([-\tilde{\tau}_2, 0], \mathbb{R}^3_+)$.

Define a mapping $F : \mathbb{R} \to \mathcal{L}(C, \mathbb{R}^3)$ and a continuous 3×3 matrix function by

$$F(t)\begin{pmatrix}\psi_1\\ \psi_2\\ \psi_3\end{pmatrix} = \begin{pmatrix} c_{13}(t)\psi_3(-\tau_h) \\ c_{23}(t)\psi_3(-\tau_h) \\ c_{31}(t)(\psi_1(-\tau_m(t)) + \psi_2(-\tau_m(t)))\end{pmatrix}, V(t) = \begin{pmatrix} c_{11}(t) & 0 & 0 \\ 0 & c_{22}(t) & 0 \\ 0 & 0 & c_{33}(t)\end{pmatrix}.$$

Let $Z(t, k)$, $t \geq k$ be the evolution matrix of the linear periodic system $\frac{dW(t)}{dt} = -V(t)W(t)$, which satisfies

$$\frac{\partial}{\partial t}Z(t, k) = -V(t)Z(t, k), \quad \forall t \geq k, \quad \text{and} \quad Z(k, k) = I, \quad \forall k \in \mathbb{R},$$

where I is the 3×3 identity matrix.

According to Zhao [36], we define the next generation operator L on C_ω:

$$[Lu](t) = \int_0^\infty Z(t, t - k)F(t - k)u(t - k + \cdot)dk, \quad \forall t \in \mathbb{R}, u \in C_\omega,$$

and the basic reproduction number

$$\mathcal{R}_0 := r(\mathrm{L}), \tag{3.6}$$

where $r(\mathrm{L})$ is the spectral radius of L. Let $\tilde{\mathcal{P}} := \tilde{\mathcal{P}}(\omega)$ be the Poincaré (period) map for system (3.5). By virtue of [36, Theorem 2.1], we obtain the following observation.

Lemma 3.5. *$\mathcal{R}_0 - 1$ has the same sign as $r(\tilde{\mathcal{P}}) - 1$, where $r(\tilde{\mathcal{P}})$ is the spectral radius of $\tilde{\mathcal{P}}$.*

Let $\tilde{\tau} = \max\{\tau_h, \max_{0\le t\le\omega}\tau_w(t), \max_{0\le t\le\omega}\tau_m(t)\}$ and

$$\begin{aligned}
\mathcal{E}(t) :&= \Big\{\upsilon \in C\big([-\tilde{\tau}, 0], \mathbb{R}^8_+\big) : \upsilon_4(s) \le K_d(t+s), \quad \forall s \in [-\tilde{\tau}, 0],\\
\upsilon_7(0) &= \int_{-\tau_m(0)}^{0} \beta_m(s)(\upsilon_2(s)+\upsilon_3(s))\upsilon_6(s)e^{-\int_s^0 \mu_m(\xi)d\xi}ds\Big\}.
\end{aligned}$$

Lemma 3.6. *For any $\upsilon \in \mathcal{E}(0)$, system (2.5) has a unique nonnegative solution $y(t,\upsilon)$ with $y_0 = \upsilon$, and $y_t(\upsilon) := (y_{1t}(\upsilon), y_{2t}(\upsilon), y_{3t}(\upsilon), y_4(t,\upsilon), y_{5t}(\upsilon), y_{6t}(\upsilon), y_7(t,\upsilon), y_{8t}(\upsilon)) \in \mathcal{E}(t)$ on $[0,+\infty)$.*

Proof. For any $\upsilon = (\upsilon_1, \upsilon_2, \upsilon_3, \upsilon_4, \upsilon_5, \upsilon_6, \upsilon_7, \upsilon_8) \in \mathcal{E}(0)$, define $H(t,\upsilon) = (H_1, H_2, H_3, H_4, H_5, H_6, H_7, H_8)^T$, where

$$\begin{aligned}
H_1 &= \Lambda_h - \beta_h(t)\upsilon_1(0)\upsilon_8(0) - \mu_h\upsilon_1(0),\\
H_2 &= \rho\beta_h(t-\tau_h)\upsilon_1(-\tau_h)\upsilon_8(-\tau_h)e^{-\mu_h\tau_h} - (q_1+\mu_h)\upsilon_2(0),\\
H_3 &= (1-\rho)\beta_h(t-\tau_h)\upsilon_1(-\tau_h)\upsilon_8(-\tau_h)e^{-\mu_h\tau_h} - (\eta_h+q_2+\mu_h)\upsilon_3(0),\\
H_4 &= \mu_b(t)\left(1-\frac{\upsilon_4(0)}{K_d(t)}\right)(\upsilon_6(0)+\upsilon_7(0)+\upsilon_8(0)) - (\kappa_d(t)+\mu_E(t))\upsilon_4(0),\\
H_5 &= \kappa_d(t)\upsilon_4(0) - (\eta_w(t)+\mu_E(t))\upsilon_5(0),\\
H_6 &= (1-\tau_w'(t))\eta_w(t-\tau_w(t))\upsilon_4(-\tau_w(t))e^{-\int_{t-\tau_w(t)}^{t}\mu_L(\xi)d\xi} - \beta_m(t)(\upsilon_2(0)+\upsilon_3(0))\upsilon_6(0)\\
&\quad -\mu_m(t)\upsilon_6(0),\\
H_7 &= \beta_m(t)(\upsilon_2(0)+\upsilon_3(0))\upsilon_6(0) - \mu_m(t)\upsilon_7(0)\\
&\quad -(1-\tau_m'(t))\beta_m(t-\tau_m(t))(\upsilon_2(-\tau_m(t))+\upsilon_3(-\tau_m(t)))\upsilon_6(-\tau_m(t))e^{-\int_{t-\tau_m(t)}^{t}\mu_m(\xi)d\xi},\\
H_8 &= (1-\tau_m'(t))\beta_m(t-\tau_m(t))(\upsilon_2(-\tau_m(t))+\upsilon_3(-\tau_m(t)))\upsilon_6(-\tau_m(t))e^{-\int_{t-\tau_m(t)}^{t}\mu_m(\xi)d\xi}\\
&\quad -\mu_m(t)\upsilon_8(0).
\end{aligned}$$

Clearly, H is continuous in $(t,\upsilon) \in \mathbb{R}\times\mathcal{E}(0)$ and Lipschitz in υ on each compact subset of $\mathcal{E}(0)$. It follows that system (2.5) has a unique solution $y(t,\upsilon)$ with $y_0 = \upsilon$ on $[0,\delta_\upsilon)$, where $\delta_\upsilon < \infty$.

For a given $\upsilon \in \mathcal{E}(0)$, when $\upsilon_i(0) = 0$ for some $i = \{1,2,3,4,5,6,8\}$, $H_i(t,\upsilon) \ge 0$; when $\upsilon_4(0) = K_d(t)$, $H_4(t,\upsilon) \le 0$. In line with [40, Theorem 5.2.1] and the compatibility condition (2.3), combining

$$E_m(t) = \int_{t-\tau_m(t)}^{t} \beta_m(s)(A_h(s)+I_h(s))S_m(s)e^{-\int_s^t \mu_m(\xi)d\xi}ds,$$

we see that for any $\upsilon \in \mathcal{E}(0)$, the unique solution $y(t,\upsilon)$ of system (2.5) with $y_0 = \upsilon$ satisfies $y_t(\upsilon) \in C([-\tilde{\tau},0],\mathbb{R}^3)\times\mathbb{R}_+\times C([-\tilde{\tau},0],\mathbb{R}^2_+)\times\mathbb{R}_+\times C([-\tilde{\tau},0],\mathbb{R}_+)$ for all $t \in [0,\delta_\upsilon)$.

We still need to prove that $y_4(t) \le K_d(t)$ for all $[0,\delta_\upsilon)$, that is, if $E_d(0) \le K_d(0)$, then $E_d(t) \le K_d(t)$ holds for all $[0,\omega)$. This inequality is true if condition (2.4) is satisfied.

Let

$$\begin{aligned}
\mathbb{F}(t) :&= \Big\{\vartheta \in C\Big([-\tilde{\tau},0],\mathbb{R}^{12}_+\Big) : \vartheta_7(s) \le K_d(t+s), \quad \forall s \in [-\tilde{\tau},0],\\
\vartheta_2(0) &= \int_{-\tau_h}^{0} \beta_h(s)\vartheta_1(s)\vartheta_{12}(s)e^{-\mu_h s}ds,\\
\vartheta_9(0) &= \int_{-\tau_w(0)}^{0} \eta_w(s)\vartheta_8(s)e^{-\int_s^0\mu_E(\xi)d\xi}ds,\\
\vartheta_{11}(0) &= \int_{-\tau_m(0)}^{0} \beta_m(s)(\vartheta_3(s)+\vartheta_4)\vartheta_{10}(s)e^{-\int_s^0\mu_m(\xi)d\xi}ds\Big\}.
\end{aligned}$$

Then the nonnegative solution $z(t,\vartheta) = (S_h(t), E_h(t), A_h(t), I_h(t), P_h(t), R_h(t), E_d(t), E_w(t), L_m(t), S_m(t), E_m(t), I_m(t))$ of system (2.2) exists uniquely and satisfies $z_0 = \vartheta$, $\forall\vartheta \in \mathbb{F}(0)$. In model (2.2), letting $N_h(t) = S_h(t) + E_h(t) + A_h(t) + I_h(t) +$

$P_h(t)+R_h(t)$ and $N_m(t)=S_m(t)+E_m(t)+I_m(t)$, we then obtain the following system:

$$\begin{cases} \frac{dN_h}{dt}(t)=\Lambda_h-\mu_h N_h(t),\\ \frac{dE_d}{dt}(t)=\mu_b(t)\big(1-\frac{E_d(t)}{K_d(t)}\big)S_m(t)-(\kappa_d(t)+\mu_E(t))E_d(t),\\ \frac{dE_w}{dt}(t)=\kappa_d(t)E_d(t)-(\eta_w(t)+\mu_E(t))E_w(t),\\ \frac{dI_m}{dt}(t)=\eta_w(t)E_w(t)-(1-\tau_w'(t))\eta_w(t-\tau_w(t))E_w(t-\tau_w(t))e^{-\int_{t-\tau_w(t)}^{t}\mu_L(\xi)d\xi}\\ \quad -\mu_L(t)L(t),\\ \frac{dN_m}{dt}(t)=(1-\tau_w'(t))\eta_w(t-\tau_w(t))E_w(t-\tau_w(t))e^{-\int_{t-\tau_w(t)}^{t}\mu_L(\xi)d\xi}-\mu_m(t)N_m(t). \end{cases} \tag{3.7}$$

From systems (3.4) and (3.7), it is obvious that $(z_1(t,\vartheta), z_2(t,\vartheta), z_3(t,\vartheta), z_4(t,\vartheta), z_5(t,\vartheta), z_6(t,\vartheta))\le \Lambda_h/\mu_h$. By Lemma 3.2, there exists a positive constant $\mathbb{K}$ such that $(z_7(t,\vartheta), z_8(t,\vartheta), z_9(t,\vartheta), z_{10}(t,\vartheta), z_{11}(t,\vartheta), z_{12}(t,\vartheta))\le \mathbb{K}$ for $t\in[0,\delta_\upsilon)$. Hence, [41, Theorem 2.3.1] implies that $\delta_\upsilon=\infty$. By virtue of the comparison argument and the global attractivity of $(S_h^*, E_d^*(t), E_w^*(t), S_m^*(t))$ for system (3.4), it follows that the solutions for system (2.2) are in $\mathbb{F}(t)$. Therefore, solutions of system (2.5) in $\mathcal{E}(t)$ exist globally on $[0,+\infty)$ and are ultimately bounded. □

Define

$$\mathcal{J}:=C([-\tau_m(0),0],\mathbb{R}^2)\times C([-\tau_h,0],\mathbb{R}),\ \mathcal{J}^+:=C([-\tau_m(0),0],\mathbb{R}_+^2)\times C([-\tau_h,0],\mathbb{R}_+).$$

To discuss the global dynamics of system (2.5), we first prove that the linear system (3.5) generates a periodic semiflow on $\mathcal{J}$, which is eventually strongly monotone.

Lemma 3.7. *For any $\varsigma=(\varsigma_1,\varsigma_2,\varsigma_3)\in\mathcal{J}^+$, the nonnegative solution $x(t,\varsigma)$ with $x_0=\varsigma$ of system (3.5) exists uniquely on* $[0,\infty)$.

The above lemma can be easily proved by defining $\tilde{\tau}_2=\min\{\tau_h,\min_{0\le t\le\omega}\tau_m(t)\}$ and the arguments similar to those in Lemma 3.3.

Remark 3.2. In line with the uniqueness of solution in Lemmas 3.6 and 3.7, we see that for any $\vartheta\in C^+$ and $\varsigma\in\mathcal{J}^+$ with $\vartheta_1(\theta_1)=\varsigma_1(\theta_1)$, $\vartheta_2(\theta_1)=\varsigma_2(\theta_1)$, $\forall\theta_1\in[-\tau_m(0),0)$, and $\vartheta_3(\theta_2)=\varsigma_3(\theta_2)$, $\forall\theta_2\in[-\tau_h,0)$, there holds $y(t,\vartheta)=z(t,\varsigma)$ for all $t\ge0$, where solutions $y(t,\vartheta)$ and $z(t,\varsigma)$ of system (3.5) satisfy $y_0=\vartheta$ and $z_0=\varsigma$, respectively.

Let $\mathcal{P}(t)$ be the solution maps of system (3.5) on $\mathcal{J}$, namely, $\mathcal{P}(t)\varsigma=x_t(\varsigma), t\ge0$, where the unique solution $x(t,\varsigma)$ of system (3.5) satisfies $x_0=\varsigma\in\mathcal{J}$. It follows from [37, Lemma 3.5] that $\mathcal{P}(t)$ is an ω-periodic semiflow on $\mathcal{J}$. Let $r(\mathcal{P})$ be the spectral radius of poincaré map $\mathcal{P}:=\mathcal{P}(\omega)$ on $\mathcal{J}$. Obviously, Lemma 3.7 implies that $\mathcal{J}^+$ is positively invariant for system (3.5) and $\mathcal{P}(t)$ is monotone on $\mathcal{J}$ for $t\ge0$. Next we show that $\mathcal{P}(t)$ is eventually strongly monotone.

Lemma 3.8. *For any $\vartheta,\zeta\in\mathcal{J}$ with $\vartheta>\zeta$, the solutions $y(t)$ and $z(t)$ of system (3.5) with $y_0=\zeta$ and $z_0=\zeta$, respectively, satisfy $y(t)\gg z(t)$ for $t\ge\tilde{\tau}_2$. Thus, $\mathcal{P}(t)\vartheta\gg\mathcal{P}(t)\zeta$ for all $t\ge4\tilde{\tau}_2$.*

Proof. Let $y(t,\vartheta)=(\hat{A}_h(t),\hat{I}_h(t),\hat{I}_m(t))$ and $z(t,\zeta)=(A_h(t),I_h(t),I_m(t))$. Without loss of generality, we assume that $\vartheta_3>\zeta_3$.

Claim 1. There exists $t_0\in[0,\tilde{\tau}_2]$ such that $\hat{A}_h(t)>A_h(t)$, $\forall t\ge t_0$.

Firstly, we prove that $\hat{A}_h(t_0)>A_h(t_0)$ for some $t_0\in[0,\tilde{\tau}_2]$. Otherwise, $\hat{A}_h(t)=A_h(t)$ holds, $\forall t\in[0,\tilde{\tau}_2]$, then $\frac{d\hat{A}_h(t)}{dt}=\frac{dA_h(t)}{dt}$, $\forall\in(0,\tilde{\tau}_2)$. Hence, we get

$$c_{13}(t)(\hat{I}_m(t-\tau_h)-I_m(t-\tau_h))=0,\ \forall t\in[0,\tilde{\tau}_2].$$

Since $-\tau_h\le0=\tilde{\tau}_2-\tilde{\tau}_2$, we obtain $\vartheta(\theta)=\zeta(\theta)$, $\forall\theta\in[-\tau_h,0]$, which contradicts the above assumption.

Let $h_1^*(t,x):=c_{13}(t)I_m(t-\tau_h)-c_{11}(t)x$. Since

$$\begin{aligned}\frac{d\hat{A}_h(t)}{dt}&=c_{13}(t)\hat{I}_m(t-\tau_h)-c_{11}(t)\hat{A}_h(t)\\&\ge c_{13}(t)I_m(t-\tau_h)-c_{11}(t)\hat{A}_h(t)\\&=h_1^*(t,\hat{A}_h(t)),\end{aligned}$$

we have $\frac{d\hat{A}_h(t)}{dt}-h_1^*(t,\hat{A}_h(t))\ge0=\frac{dA_h(t)}{dt}-h_1^*(t,A_h(t)),\forall t\ge t_0$. It follows from the comparison theorem [38, Theorem 4] and the inequality $\hat{A}_h(t_0)>A_h(t_0)$ that $\hat{A}_h(t)>A_h(t)$, $\forall t\ge t_0$.

Claim 2. Let $I(t)=A_h(t)+I_h(t)$ and $\hat{I}(t)=\hat{A}_h(t)+\hat{I}_h(t)$. Then there exists $t_1\in[0,\tilde{\tau}_2]$ such that $\hat{I}(t)>I(t)$, $\forall t\ge t_1$.

By the argument similar to that in Claim 1, we can prove the following claim.

Claim 3. $\hat{I}_m(t)>I_m(t)$, $\forall t\ge t_1+\tilde{\tau}_2$.

Let $h_2^*(t,x):=c_{31}(t)(A_h(t-\tau_m(t))+I_h(t-\tau_m(t)))-c_{33}(t)x$. Since

$$\begin{aligned}\frac{d\hat{I}_m(t)}{dt}&=c_{31}(t)(\hat{A}_h(t-\tau_m(t))+\hat{I}_h(t-\tau_m(t)))-c_{33}(t)\hat{I}_m(t)\\&\ge c_{31}(t)(A_h(t-\tau_m(t))+I_h(t-\tau_m(t)))-c_{33}(t)\hat{I}_m(t)\end{aligned}$$

$$= h_2^*(t, \hat{I}_m(t)),$$

we have $\frac{d\hat{I}_m(t)}{dt} - h_2^*(t, \hat{I}_m(t)) \geq 0 = \frac{dI_m(t)}{dt} - h_2^*(t, I_m(t)), \forall t \geq t_1 + \tilde{\tau}_2$. Then it follows from the comparison theorem [38, Theorem 4] and the inequality $\hat{I}_m(t_1 + \tilde{\tau}_2) > I_m(t_1 + \tilde{\tau}_2)$ that $\hat{I}_m(t) > I_m(t)$, $\forall t \geq t_1 + \tilde{\tau}_2$.

Claim 4. $\hat{I}_h(t) > I_h(t)$, $\forall t \geq t_0 + t_1 + 2\tilde{\tau}_2$.

The above claim can be proved by an argument similar to that in Claim 3. Since $t_0 \in [0, \tilde{\tau}_2]$ and $t_1 \in [0, \tilde{\tau}_2]$, we then have $\mathcal{P}(t)\vartheta \gg \mathcal{P}(t)\zeta$ for $t \geq 4\tilde{\tau}_2$. □

By virtue of Lemma 3.8 and the argument in [37, Lemma 3.5], we have the following observation.

Lemma 3.9. *$r(\tilde{\mathcal{P}}) = r(\mathcal{P})$, and hence, $\mathcal{R}_0 - 1$ has the same sign as $r(\mathcal{P}) - 1$, where $\mathcal{R}_0$ is defined in (3.6).*

Let

$$\begin{aligned}\mathbb{M}(t) :=& C([-\tau_h, 0], \mathbb{R}_+) \times C([-\tau_m(0), 0], \mathbb{R}_+^2) \times [0, K_d(t)] \times C([-\tau_w(0), 0], \mathbb{R}_+)\\ &\times C([-\tau_m(0), 0], \mathbb{R}_+) \times \mathbb{R}_+ \times C([-\tau_h, 0], \mathbb{R}_+), \quad \forall t \geq 0,\\ \mathcal{M}(t) :=& \left\{\psi \in \mathbb{M}(t) : \psi_7(0) = \int_{-\tau_m(0)}^0 \beta_m(s)(\psi_2(s) + \psi_3(s))\psi_6(s)e^{-\int_s^0 \mu_m(\xi)d\xi} ds\right\}.\end{aligned}$$

As discussed in Lemma 3.7, we see that for any $\upsilon \in \mathcal{E}(0)$ and $\psi \in \mathcal{M}(0)$ with $\upsilon_i(\theta_i) = \psi_i(\theta_i)$, $\forall \theta_i \in [-\tau_h, 0), i = 1, 8$, $\upsilon_j(\theta_j) = \psi_j(\theta_j)$, $\forall \theta_j \in [-\tau_m(0), 0), j = 2, 3, 6$, $\upsilon_k = \psi_k$, $\forall k = 4, 7$, and $\upsilon_5(\theta_5) = \psi_5(\theta_5)$, $\forall \theta_5 \in [-\tau_w(0), 0)$, there holds $u(t, \upsilon) = v(t, \psi)$ for all $t \geq 0$, where the solutions $u(t, \upsilon)$ and $v(t, \psi)$ of system (2.5) satisfy $u_0 = \upsilon$ and $v_0 = \psi$, respectively.

Now we are in a positive to prove the main results of this section.

Theorem 3.1. *Let $\mathcal{R}_m$ and $\mathcal{R}_0$ be defined as in (3.3) and (3.6), respectively. Then the following statements hold true:*

(i) If $\mathcal{R}_m \leq 1$, then $E_0 = (S_h^*, 0, 0, 0, 0, 0, 0, 0)$ is globally attractive for system (2.5) in $\mathcal{M}(0)$;

(ii) If $\mathcal{R}_m > 1$ and $\mathcal{R}_0 < 1$, then the disease-free ω-periodic solution $(S_h^*, 0, 0, E_d^*(t), E_w^*(t), S_m^*(t), 0, 0)$ of system (2.5) is globally attractive in $\mathcal{M}(0) \setminus \{E_0\}$;

(iii) If $\mathcal{R}_m > 1$ and $\mathcal{R}_0 > 1$, then there exists $\sigma > 0$ such that for any $\upsilon \in \mathcal{M}(0)$ with $\upsilon_2(0) > 0$, $\upsilon_3(0) > 0$ and $\upsilon_8(0) > 0$, the solution $x(t, \upsilon) = (S_h(t), A_h(t), I_h(t), E_d(t), E_w(t), S_m(t), E_m(t), I_m(t))$ of system (2.5) satisfies

$$\liminf_{t\to\infty}(A_h(t), I_h(t), I_m(t)) \geq (\sigma, \sigma, \sigma).$$

Proof. Let $G(t)$ be the solution maps of system (2.5). By the arguments similar to [37, Lemma 3.5], it follows that $G(t) : \mathcal{M}(0) \to \mathcal{M}(t)$ is an ω-periodic semiflow of system (2.5). Thus, $G := G(\omega)$ is the Poincaré map on $\mathcal{M}(0)$, and $\{G^n\}_{n\geq 0}$ is a discrete-time system on $\mathcal{M}(0)$. For any given $\psi \in \mathcal{M}(0)$, the omega limit set of the orbit $\{G^n(\psi)\}_{n\geq 0}$ is $\mathfrak{K}(\psi)$. From [39, Lemma 1.2.1], $\mathfrak{K}(\psi)$ is an internally chain transitive set for $\{G^n\}$ on $\mathcal{M}(0)$.

Since the proofs for case (i) and (ii) are similar, we only prove the latter.

(ii) If $\mathcal{R}_m > 1$ and $\mathcal{R}_0 < 1$, in view of Lemma 3.9 and [42, Lemma 1], then we have $r(\mathcal{P}) < 1$ and $\mu = \ln r(\mathcal{P})/\omega < 0$. Define a perturbation system on $\mathcal{J}$ as follows:

$$\begin{cases}\frac{dA_h}{dt}(t) = \rho\beta_h(t-\tau_h)(S_h^* + \varepsilon)I_m(t-\tau_h)e^{-\mu_h\tau_h} - (q_1 + \mu_h)A_h(t),\\ \frac{dI_h}{dt}(t) = (1-\rho)\beta_h(t-\tau_h)(S_h^* + \varepsilon)I_m(t-\tau_h)e^{-\mu_h\tau_h} - (\eta_h + q_2 + \mu_h)I_h(t),\\ \frac{dI_m}{dt}(t) = (1-\tau_m'(t))\beta_m(t-\tau_m(t))(A_h(t-\tau_m(t)) + I_h(t-\tau_m(t)))\\ \quad\times(S_m^*(t-\tau_m(t)) + \varepsilon)e^{-\int_{t-\tau_m(t)}^t \mu_m(\xi)d\xi} - \mu_m(t)I_m(t).\end{cases} \tag{3.8}$$

Let $\mathcal{P}_\varepsilon(t)$ be the solution maps of system (3.8) and $\mathcal{P}_\varepsilon = \mathcal{P}_\varepsilon(\omega)$. Since $\lim_{\varepsilon\to 0} r(\mathcal{P}_\varepsilon) = r(\mathcal{P}) < 1$, we can choose a positive number ε small enough for $r(\mathcal{P}_\varepsilon) < 1$. Then it is easy to prove the compactness and eventual strong monotonicity of $\mathcal{P}_\varepsilon(t)$ on $\mathcal{J}$ for $t \geq 4\hat{\tau}$. By [42, Lemma 1], there exists a positive ω-periodic function $v_\varepsilon(t)$ such that $u_\varepsilon(t) = e^{\mu_\varepsilon t}v_\varepsilon(t)$ is a positive solution of system (3.8), where $\mu_\varepsilon = \ln r(\mathcal{P}_\varepsilon)/\omega$, and hence, $\lim_{t\to\infty} u_\varepsilon(t) = 0$.

From system (3.4) and the global attractivity of S_h^* and $S_m^*(t)$, it follows that there exists a positive integer N_1 large enough for $N_1\omega > \tilde{\tau}_2$ and $S_h^* - \varepsilon \leq S_h(t) \leq S_h^* + \varepsilon$, $S_m^*(t) - \varepsilon \leq S_m(t) \leq S_m^*(t) + \varepsilon$, $\forall t \geq N_1\omega - \tilde{\tau}_2$. Then we obtain

$$\begin{cases}\frac{dA_h}{dt}(t) = \rho\beta_h(t-\tau_h)S_h(t-\tau_h)I_m(t-\tau_h)e^{-\mu_h\tau_h} - (q_1 + \mu_h)A_h(t)\\ \leq \rho\beta_h(t-\tau_h)(S_h^* + \varepsilon)I_m(t-\tau_h)e^{-\mu_h\tau_h} - (q_1 + \mu_h)A_h(t),\\ \frac{dI_h}{dt}(t) = (1-\rho)\beta_h(t-\tau_h)S_h(t-\tau_h)I_m(t-\tau_h)e^{-\mu_h\tau_h} - (\eta_h + q_2 + \mu_h)I_h(t)\\ \leq (1-\rho)\beta_h(t-\tau_h)(S_h^* + \varepsilon)I_m(t-\tau_h)e^{-\mu_h\tau_h} - (\eta_h + q_2 + \mu_h)I_h(t),\\ \frac{dI_m}{dt}(t) = (1-\tau_m'(t))\beta_m(t-\tau_m(t))(A_h(t-\tau_m(t)) + I_h(t-\tau_m(t)))S_m(t-\tau_m(t))\\ \quad\times e^{-\int_{t-\tau_m(t)}^t \mu_m(\xi)d\xi} - \mu_m(t)I_m(t)\\ \leq (1-\tau_m'(t))\beta_m(t-\tau_m(t))(A_h(t-\tau_m(t)) + I_h(t-\tau_m(t)))(S_m^*(t-\tau_m(t)) + \varepsilon)\\ \quad\times e^{-\int_{t-\tau_m(t)}^t \mu_m(\xi)d\xi} - \mu_m(t)I_m(t)\end{cases}$$

for all $t \geq N_1\omega$. Choose a large enough positive number ν such that $(A_h(t), I_h(t), I_m(t)) \leq \nu\mu_\varepsilon(t)$, $\forall t \in [N_1\omega, N_1\omega + \tilde{\tau}_2]$. It follows from [40, Theorem 5.1.1] that $(A_h(t), I_h(t), I_m(t)) \leq \nu\mu_\varepsilon(t)$, $\forall t \geq N_1\omega + \tilde{\tau}_2$. Thus, $\lim_{t\to\infty} A_h(t) = \lim_{t\to\infty} I_h(t) = \lim_{t\to\infty} I_m(t) = 0$.

Next we can prove that

$$\lim_{t\to\infty} \mid (S_h(t), A_h(t), I_h(t), E_d(t), E_w(t), S_m(t), E_m(t), I_m(t)) - (S_h^*, 0, 0, E_d^*(t), E_w^*(t), S_m^*(t), 0, 0) \mid = 0$$

by using the theory of internally chain transitive sets (see [39, Section 1.2]).

Since $\mathcal{R}_m > 1$, Lemma 3.1 (ii) implies that

$$\lim_{n\to\infty} ((Q^n(\psi))_1, (Q^n(\psi))_4, (Q^n(\psi))_5, (Q^n(\psi))_6) = (S_h^*, E_d^*(0), E_{w0}^*, S_{m0}^*),$$

where $E_{w0}^* = E_w^*(\theta)$, $\forall \theta \in [-\tau_w(0), 0]$, $S_{m0}^* = S_m^*(\theta)$, $\forall \theta \in [-\tau_m(0), 0]$. Thus, there exists a subset $\mathfrak{K}_1$ of $C([-\tau_m(0), 0], \mathbb{R}_+^2) \times \mathbb{R}_+ \times C([-\tau_h, 0], \mathbb{R}_+)$ such that

$$\mathfrak{K}(\psi) = \{(S_h^*, E_d^*(0), E_{w0}^*, S_{m0}^*)\} \times \mathfrak{K}_1.$$

For any given $\tilde{k} = (\tilde{k}_1, \tilde{k}_2, \tilde{k}_3, \tilde{k}_4, \tilde{k}_5, \tilde{k}_6, \tilde{k}_7, \tilde{k}_8) \in \mathfrak{K}(\psi)$, there exists a sequence $n_i \to \infty$ such that $G^{n_i}(\psi) \to \tilde{k}$ as $i \to \infty$. Since $0 \le (G^{n_i}(\psi))_2 = A_h(n_i\omega, \psi)$, $0 \le (G^{n_i}(\psi))_3 = I_h(n_i\omega, \psi)$, $0 \le (G^{n_i}(\psi))_7 = E_m(n_i\omega, \psi)$, and $0 \le (G^{n_i}(\psi))_8 = I_m(n_i\omega, \psi)$ for all $\psi \in \mathcal{M}(0)$, letting $n_i \to \infty$, we then have $0 \le (\tilde{k}_2, \tilde{k}_3, \tilde{k}_7, \tilde{k}_8)$. It is clear that

$$G^n \mid_{\mathfrak{K}(\psi)} (S_h^*, \tilde{k}_2, \tilde{k}_3, E_d^*(0), E_{w0}^*, S_{m0}^*, \tilde{k}_7, \tilde{k}_8) = \{(S_h^*, E_d^*(0), E_{w0}^*, S_{m0}^*)\} \times G_1^n \mid_{\mathfrak{K}_1} (\tilde{k}_2, \tilde{k}_3, \tilde{k}_7, \tilde{k}_8),$$

and $\mathfrak{K}_1$ is an internally chain transitive set for $\{G_1^n\}$. According to the aforementioned discussion, when $r(\mathcal{P}) < 1$, $\lim_{n\to\infty} ((G^n(\psi))_2, G^n(\psi))_3, (G^n(\psi))_8) = (0, 0, 0)$. By virtue of [39, Theorem 1.2.1], we obtain that $\mathfrak{K}_1 = \{(0, 0, 0, 0)\}$, and thus, $\mathfrak{K}(\psi) = \{(S_h^*, 0, 0, E_d^*(0), E_{w0}^*, S_{m0}^*, 0, 0)\}$.

(iii) If $\mathcal{R}_m > 1$ and $\mathcal{R}_0 > 1$, by Lemma 3.9 and [42, Lemma 1], then we have $r(\mathcal{P}) > 1$ and $\mu = \ln r(\mathcal{P})/\omega > 0$. Define

$$\mathcal{M}_0 := \{\nu = (\nu_1, \nu_2, \nu_3, \nu_4, \nu_5, \nu_6, \nu_7, \nu_8) \in \mathcal{M}(0) : \nu_i(0) > 0, i = 2, 3, 8\},$$
$$\partial\mathcal{M}_0 := \mathcal{M}(0) \setminus \mathcal{M}_0 = \{\nu \in \mathcal{M}(0) : \nu_i(0) = 0, \text{ for some } i \in \{2, 3, 8\}\}.$$

From the second, third and eighth equations of system (2.5), it follows that $G(t)\mathcal{M}_0 \subseteq \mathcal{M}_0$, $\forall t \ge 0$. According to Lemma 3.2, Remark 3.1 and [41, Theorem 3.6.1], G is compact for $t \ge \tilde{\tau}_2$, and thus, system $\{G^n : \mathcal{M}(0) \to \mathcal{M}(0)\}_{n\ge 0}$ is point dissipative and G^n is compact for large enough n. Therefore, G has a strong global attractor in $\mathcal{M}(0)$ by [39, Theorem 1.1.3]. Next we prove the uniform persistence of G with respect to $\mathcal{M}_0$.

Denote $\mathcal{N}_1 = \{(S_h^*, 0, 0, 0, 0, 0, 0, 0)\}$, $\mathcal{N}_2 = \{(S_h^*, 0, 0, E_d^*(0), E_{w0}^*, S_{m0}^*, 0, 0)\}$, where $E_{w0}^* = E_w^*(\theta)$, $\forall \theta \in [-\tau_w(0), 0]$, $S_{m0}^* = S_m^*(\theta)$, $\forall \theta \in [-\tau_m(0), 0]$.

Claim. For each $i = 1, 2$, there exists $\eta_i > 0$ such that $\limsup_{n\to\infty} \|G(n\omega)\vartheta - \mathcal{N}_i\| \ge \eta_i$, $\forall \vartheta \in \mathcal{M}_0$.

We just prove the claim for $i = 2$, and the other case can be proved in a similar way. Since $\lim_{\vartheta\to\mathcal{N}_2} \|G(t)\vartheta - G(t)\mathcal{N}_2\| = 0$ uniformly, $\forall t \in [0, \omega]$, for any given $\varrho > 0$, there is $\eta_2 > 0$ such that for any $\vartheta \in \mathcal{M}_0$ with $\|\vartheta - \mathcal{N}_2\| < \eta_2$, we have $\|G(t)\vartheta - G(t)\mathcal{N}_2\| < \varrho$, $\forall t \in [0, \omega]$.

Assume, by contradiction, that $\limsup_{n\to\infty} \|G(n\omega)\xi - \mathcal{N}_2\| < \eta_2$ for some $\xi \in \mathcal{M}_0$. Then there exists an integer $N_2 \ge 1$ such that $\|G(n\omega)\xi - \mathcal{N}_2\| < \eta_2$, $\forall n \ge N_2$. For any $t \ge N_2\omega$, let $t = n\omega + \hat{t}$ with $n \ge N_2$ and $\hat{t} \in [0, \omega]$, then $\|G(t)\xi - G(t)\mathcal{N}_2\| = \|G(\hat{t})G(n\omega)\xi - G(\hat{t})G(n\omega)\mathcal{N}_2\| = \|G(\hat{t})G(n\omega)\xi - G(\hat{t})\mathcal{N}_2\| < \varrho$, $\forall t \ge N_2\omega$. Thus, $S_h(t-\tau) > S_h^* - \varrho$ and $S_m(t - \tau_m(t)) > S_m^*(t - \tau_m(t)) - \varrho$, $\forall t \ge N_2\omega - \tilde{\tau}_2$. Define the perturbation system as follows:

$$\begin{cases} \frac{dA_h}{dt}(t) = \rho\beta_h(t-\tau_h)(S_h^* - \varrho)I_m(t-\tau_h)e^{-\mu_h\tau_h} - (q_1+\mu_h)A_h(t), \\ \frac{dI_h}{dt}(t) = (1-\rho)\beta_h(t-\tau_h)(S_h^* - \varrho)I_m(t-\tau_h)e^{-\mu_h\tau_h} - (\eta_h+q_2+\mu_h)I_h(t), \\ \frac{dI_m}{dt}(t) = (1-\tau_m'(t))\beta_m(t-\tau_m(t))(A_h(t-\tau_m(t)) + I_h(t-\tau_m(t))) \\ \quad \times (S_m^*(t-\tau_m(t)) - \varrho)e^{-\int_{t-\tau_m(t)}^t \mu_m(\xi)d\xi} - \mu_m(t)I_m(t), \end{cases} \tag{3.9}$$

for all $t \ge N_2\omega$. Let $\mathcal{P}_\varrho$ be the Poincaré map of system (3.9). From $\lim_{\varrho\to 0} r(\mathcal{P}_\varrho) = r(\mathcal{P}) > 1$, we can choose a positive number ϱ small enough for $r(\mathcal{P}_\varrho) > 1$. By [42, Lemma 1], there exists an ω-periodic function $v_\varrho(t)$ such that $u_\varrho(t) = e^{\mu_\varrho t}v_\varrho(t)$ is a positive solution of system (3.9), where $\mu_\varrho = \ln r(\mathcal{P}_\varrho)/\omega$. Obviously, $\lim_{t\to\infty} u_\varrho(t) = \infty$. For $t \ge N_2\omega - \tilde{\tau}_2$, we have

$$\begin{cases} \frac{dA_h}{dt}(t) = \rho\beta_h(t-\tau_h)S_h(t-\tau_h)I_m(t-\tau_h)e^{-\mu_h\tau_h} - (q_1+\mu_h)A_h(t) \\ \ge \rho\beta_h(t-\tau_h)(S_h^* - \varrho)I_m(t-\tau_h)e^{-\mu_h\tau_h} - (q_1+\mu_h)A_h(t), \\ \frac{dI_h}{dt}(t) = (1-\rho)\beta_h(t-\tau_h)S_h(t-\tau_h)I_m(t-\tau_h)e^{-\mu_h\tau_h} - (\eta_h+q_2+\mu_h)I_h(t) \\ \ge (1-\rho)\beta_h(t-\tau_h)(S_h^* - \varrho)I_m(t-\tau_h)e^{-\mu_h\tau_h} - (\eta_h+q_2+\mu_h)I_h(t), \\ \frac{dI_m}{dt}(t) = (1-\tau_m'(t))\beta_m(t-\tau_m(t))(A_h(t-\tau_m(t)) + I_h(t-\tau_m(t)))S_m(t-\tau_m(t)) \\ \quad \times e^{-\int_{t-\tau_m(t)}^t \mu_m(\xi)d\xi} - \mu_m(t)I_m(t) \\ \ge (1-\tau_m'(t))\beta_m(t-\tau_m(t))(A_h(t-\tau_m(t)) + I_h(t-\tau_m(t))) \\ \quad \times (S_m^*(t-\tau_m(t)) - \varrho)e^{-\int_{t-\tau_m(t)}^t \mu_m(\xi)d\xi} - \mu_m(t)I_m(t). \end{cases}$$

Since $A_h(t, \xi) > 0$, $I_h(t, \xi) > 0$, $I_m(t, \xi) > 0$, $\forall t \ge 0$, we can choose $\phi > 0$ large enough such that $(A_h(t), I_h(t), I_m(t)) \ge \phi\mu_\varrho(t)$, $\forall t \in [N_2\omega + \tilde{\tau}_2, N_2\omega + 2\tilde{\tau}_2]$. With the comparison theorem [40, Theorem 5.1.1], $(A_h(t), I_h(t), I_m(t)) \ge \phi\mu_\varrho(t)$ holds, $\forall t \ge N_2\omega + 2\tilde{\tau}_2$. Thus, $\lim_{t\to\infty} A_h(t, \xi) = \lim_{t\to\infty} I_h(t, \xi) = \lim_{t\to\infty} I_m(t, \xi) = \infty$, which is a contradiction.

Let

$$\tilde{\mathcal{N}}_\partial := \{\varphi \in \partial\mathcal{M}_0 : G^n(\varphi) \in \partial\mathcal{M}_0, \forall n \ge 0\}.$$

$$Y_1 := \{\varphi \in \mathcal{M}(0) : \varphi_i(0) = 0, \forall i = \{2, 3, 8\}\}.$$

$$Y_2 := \{\varphi \in \mathcal{M}(0) : \varphi_i(0) = 0, \forall i = \{4, 5, 6, 7, 8\}\}.$$

When (0,0,0) is an equilibrium of system (3.1), we claim that $\widetilde{\mathcal{N}}_\partial = Y_1 \cup Y_2$ holds. We first prove that $Y_1 \cup Y_2 \subset \widetilde{\mathcal{N}}_\partial$. For any $\upsilon \in Y_2$, it can easily be seen that $u_i(t, \upsilon) = 0$ for $i = 4, 5, 6, 7, 8$. Therefore, $Y_2 \subset \widetilde{\mathcal{N}}_\partial$. For any $\upsilon \in Y_1$, we can define $U(t) \in C(\mathbb{R}_+, \mathbb{R}^8_+)$ such that $U_i(t, \upsilon) = 0$ for $i = 2, 3, 7, 8$. Let $U_i(t)$ $(i = 1, 4, 5, 6)$ satisfy the following equations

$$\begin{cases} \frac{dU_1}{dt}(t) = \Lambda_h - \mu_h U_1, \quad t \geq 0 \quad \text{with} \quad U_1(0) = \upsilon_1(0), \\ \frac{dU_4}{dt}(t) = \mu_b(t)\big(1 - \frac{U_4(t)}{K_d(t)}\big)U_6(t) - (\kappa_d(t) + \mu_E(t))U_4(t), \quad t \geq 0, \\ \frac{dU_5}{dt}(t) = \kappa_d(t)U_4(t) - (\eta_w(t) + \mu_E(t))U_5(t), \quad t \geq 0, \\ \frac{dU_6}{dt}(t) = (1 - \tau_w'(t))\eta_w(t - \tau_w(t))\upsilon_5(t - \tau_w(t))e^{-\int_{t-\tau_w(t)}^{t} \mu_L(\xi)d\xi} - \mu_m(t)U_6(t), \quad 0 \leq t \leq \tilde{\tau}_1, \\ \frac{dU_6}{dt}(t) = (1 - \tau_w'(t))\eta_w(t - \tau_w(t))U_5(t - \tau_w(t))e^{-\int_{t-\tau_w(t)}^{t} \mu_L(\xi)d\xi} - \mu_m(t)U_6(t), \quad t \geq \tilde{\tau}_1. \end{cases}$$

It follows from Lemma 3.2 that $U(t)$ is a solution of system (2.5) through υ. According to the uniqueness of the solution, then $u(t, \upsilon) = U(t)$ for all $t \geq 0$, where $u(t, \upsilon)$ is a solution of system (2.5), and thus $Y_1 \subset \widetilde{\mathcal{N}}_\partial$.

Next we prove $\widetilde{\mathcal{N}}_\partial \subset Y_1 \cup Y_2$. For any $\upsilon \in \partial\mathcal{M}_0 \setminus (Y_1 \cup Y_2)$, we obtain $\upsilon_4(0) > 0$, $\upsilon_5(0) > 0$, and $\sum_{j=6}^{8} \upsilon_j(0) > 0$, and therefore $\lim_{t\to\infty} |u_4(t, \upsilon) - E_d^*(t)| = \lim_{t\to\infty} |u_5(t, \upsilon) - E_w^*(t)| = \lim_{t\to\infty} |u_6(t, \upsilon) - S_m^*(t)| = 0$. By the sixth equation of system (2.5), there exists a $t_0 > 0$ such that $u_6(t, \upsilon) > 0$, $\forall t > t_0$. Clearly, $u_1(t, \upsilon) > 0$ for all $t > 0$ from the first equation of system (2.5). With the seventh and eighth of system (2.5), if $\upsilon_7(0) > 0$, then $u_7(t, \upsilon) > 0$ and $u_8(t, \upsilon) > 0$ for all $t > 0$. From the second, third and eighth equations of system (2.5), if $\upsilon_8(0) > 0$, then $u_j(t, \upsilon) > 0$ for all $j \in \{2, 3, 8\}$ and $t > 0$. Besides, if $\upsilon_2(0) > 0$ or $\upsilon_3(0) > 0$, then $u_j(t, \upsilon) > 0$ for all $j \in \{2, 3\}$ and $t > 0$. Thus, we obtain $u(t, \upsilon) \in \mathcal{M}_0$, $\forall t > t_0$, which indicates that for any $\upsilon \in \partial\mathcal{M}_0 \setminus (Y_1 \cup Y_2)$, there exists some n with $nT > t_0$ such that $G^n(\varphi) \in \partial\mathcal{M}_0$. As a result, $\widetilde{\mathcal{N}}_\partial \subset Y_1 \cup Y_2$. Therefore, $\mathcal{N}_1$ and $\mathcal{N}_2$ are disjoint, compact and isolated invariant sets for G in $\widetilde{\mathcal{N}}_\partial$, and $\bigcup_{\upsilon\in\widetilde{\mathcal{N}}_\partial} \omega(\upsilon) = \{\mathcal{N}_1, \mathcal{N}_2\}$. Further, there is no subset of $\{\mathcal{N}_1, \mathcal{N}_2\}$ that can form a cycle in $\widetilde{\mathcal{N}}_\partial$. By virtue of the aforesaid claim, it follows that $\mathcal{N}_1$ and $\mathcal{N}_2$ are isolated invariant sets for G in $\mathcal{M}(0)$, and $W^s\{\mathcal{N}_i\} \bigcap \mathcal{M}_0 = \emptyset$ for all $i = 1, 2$, where $W^s\{\mathcal{N}_i\}$ is the stable set of $\mathcal{N}_i$ for G.

When (0,0,0) is not an equilibrium of system (3.1), $\widetilde{\mathcal{N}}_\partial = Y_1$ can be proved by a similar way. Then Y_2 is the only compact invariant set for G in $\widetilde{\mathcal{N}}_\partial$, so we just need to select $i = 2$ in the above argument.

In view of the acyclicity theorem on uniform persistence for maps (see [39, Theorem 1.3.1 and Remark 1.3.1]), then $G : \mathcal{M}(0) \to \mathcal{M}(0)$ is uniformly persistent with respect to $\mathcal{M}_0$. It then follows from [43, Theorem 4.5] that system (2.5) has a T-periodic solution $\Psi(t)\varphi$ with $\varphi \in \mathcal{M}_0$.

With [43, Theorem 4.5], $G : \mathcal{M}_0 \to \mathcal{M}_0$ admits a compact global attractor $\mathbb{X}_0$. For any $\phi \in \mathbb{X}_0$, $\phi_i(0) > 0$ for each $i = 2, 3, 8$. Denote $\mathbb{Y}_0 := \bigcup_{t\in[0,\omega]} G(t)\mathbb{X}_0$, then $\nu_i(0) > 0, i = 2, 3, 8$ for $\nu \in \mathbb{Y}_0$. Moreover, $\mathbb{Y}_0 \subseteq \mathcal{M}_0$ and $\lim_{t\to\infty} d(G(t)\vartheta, \mathbb{Y}_0) = 0$, $\forall \vartheta \in \mathcal{M}_0$.

Let $g : \mathcal{M}(0) \to \mathbb{R}_+$ be a continuous function, where

$$g(\nu) = \min_{i=2,3,8}\{\nu_i(0)\}, \quad \forall \nu \in \mathcal{M}(0).$$

Since $\mathbb{Y}_0$ is a compact subset of $\mathcal{M}_0$, then $\inf_{\nu\in\mathbb{Y}_0} g(\nu) = \min_{\nu\in\mathbb{Y}_0} g(\nu) > 0$. In consequence, there exists $\sigma > 0$ such that

$$\liminf_{t\to\infty} \min(A_h(t, \nu), I_h(t, \nu), I_m(t, \nu)) = \liminf_{t\to\infty} g(G(t)\nu) \geq (\sigma, \sigma, \sigma), \quad \forall \nu \in \mathcal{M}_0.$$

This completes the proof. □

Next we discuss the dynamics for the variables $E_h(t)$, $P_h(t)$, $R_h(t)$ and $L_m(t)$ in system (2.2). Note that

$$E_h(t) = \int_{t-\tau_h}^{t} \beta_h(s)I_m(s)S_h(s)e^{-\mu_h s}ds, \quad L_m(t) = \int_{t-\tau_w(0)}^{t} \eta_w(s)E_w(s)e^{-\int_s^0 \mu_E(\xi)d\xi}ds. \tag{3.10}$$

From Eq. (3.10) and Theorem 3.1 (i), if $\mathcal{R}_m \leq 1$, then $\lim_{t\to\infty} E_h(t) = 0$. And $\lim_{t\to\infty}(P_h(t), R_h(t), L_m(t)) = (0, 0, 0)$ is easily proved by the theory of internally chain transitive sets ([39]). With the arguments similar to those above, we can prove that if $\mathcal{R}_m > 1$ and $\mathcal{R}_0 < 1$, then $\lim_{t\to\infty}(E_h(t), P_h(t), R_h(t)) = (0, 0, 0)$ and $\lim_{t\to\infty} | L_m(t) - L_m^*(t) | = 0$.

By system (2.5), Theorem 3.1 (iii) and Eq. (3.10), it follows that if $\mathcal{R}_m > 1$ and $\mathcal{R}_0 > 1$, then there exists $\sigma_1 > 0$ such that

$$\liminf_{t\to\infty}(E_h(t, \nu), P_h(t, \nu), R_h(t, \nu), L_m(t, \nu)) \geq (\sigma_1, \sigma_1, \sigma_1, \sigma_1),$$

where $E_h(0)$ and $L_m(0)$ satisfy compatibility condition (2.3).

Consequently, we have the following result about the global dynamics of system (2.2).

Theorem 3.2. *Let $\mathcal{R}_m$ and $\mathcal{R}_0$ be defined as in (3.3) and (3.6), respectively. Then the following statements hold true:*

(i) If $\mathcal{R}_m \leq 1$, then $E_0^* = (S_h^*, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)$ is globally attractive for system (2.2) in $\mathbb{F}(0)$;

(ii) If $\mathcal{R}_m > 1$ *and* $\mathcal{R}_0 < 1$, *then the disease-free* ω*-periodic solution* $(S_h^*, 0, 0, 0, 0, 0, E_d^*(t), E_w^*(t), L_m^*(t), S_m^*(t), 0, 0)$ *of system (2.2) is globally attractive in* $\mathbb{F}(0) \setminus \{E_0^*\}$;

(iii) If $\mathcal{R}_m > 1$ *and* $\mathcal{R}_0 > 1$, *then there exists* $\sigma > 0$ *such that for any* $\nu \in \mathbb{F}(0)$ *with* $\nu_3(0) > 0$, $\nu_4(0) > 0$ *and* $\nu_{12}(0) > 0$, *the solution* $x(t, \nu) = (S_h(t), E_h(t), A_h(t), I_h(t), P_h(t), R_h(t), E_d(t), E_w(t), L_m(t), S_m(t), E_m(t), I_m(t))$ *of system (2.2) satisfies*

$$\liminf_{t\to\infty}(A_h(t), I_h(t), I_m(t)) \geq (\sigma, \sigma, \sigma).$$

At the end of this section, we note that when all the periodic coefficients are constant, model (2.5) reduces to the following ODE model:

$$\begin{cases} \frac{dS_h}{dt}(t) = \Lambda_h - \beta_h S_h(t) I_m(t) - \mu_h S_h(t), \\ \frac{dA_h}{dt}(t) = \rho\beta_h S_h(t-\tau_h) I_m(t-\tau_h) e^{-\mu_h\tau_h} - (q_1+\mu_h) A_h(t), \\ \frac{dI_h}{dt}(t) = (1-\rho)\beta_h(t-\tau_h) S_h(t-\tau_h) I_m(t-\tau_h) e^{-\mu_h\tau_h} - (\eta_h + q_2 + \mu_h) I_h(t), \\ \frac{dE_d}{dt}(t) = \mu_b\Big(1 - \frac{E_d(t)}{K_d}\Big)(S_m(t) + E_m(t) + I_m(t)) - (\kappa_d + \mu_E) E_d(t), \\ \frac{dE_w}{dt}(t) = \kappa_d E_d(t) - (\eta_w + \mu_E) E_w(t), \\ \frac{dS_m}{dt}(t) = \eta_w E_w(t-\tau_w) e^{-\mu_L\tau_w} - \beta_m(A_h(t) + I_h(t)) S_m(t) - \mu_m S_m(t), \\ \frac{dE_m}{dt}(t) = \beta_m(A_h(t) + I_h(t)) S_m(t) - \mu_m E_m(t) \\ \quad -\beta_m(A_h(t-\tau_m) + I_h(t-\tau_m)) S_m(t-\tau_m) e^{-\mu_m\tau_m}, \\ \frac{dI_m}{dt}(t) = \beta_m(A_h(t-\tau_m) + I_h(t-\tau_m)) S_m(t-\tau_m) e^{-\mu_m\tau_m} - \mu_m I_m(t). \end{cases} \tag{3.11}$$

Then the matrices $\tilde{F}(t)$, $\tilde{V}(t)$, $F(t)$ and $V(t)$, respectively, become

$$\tilde{F} = \begin{pmatrix} 0 & 0 & \mu_b \\ \kappa_d & 0 & 0 \\ 0 & \eta_w e^{-\mu_L\tau_w} & 0 \end{pmatrix}, \quad \tilde{V} = \begin{pmatrix} \kappa_d + \mu_E & 0 & 0 \\ 0 & \eta_w + \mu_E & 0 \\ 0 & 0 & \mu_m \end{pmatrix},$$

$$F = \begin{pmatrix} 0 & 0 & \rho\beta_h S_h^* e^{-\mu_h\tau_h} \\ 0 & 0 & (1-\rho)\beta_h S_h^* e^{-\mu_h\tau_h} \\ \beta_m S_m^* e^{-\mu_m\tau_m} & \beta_m S_m^* e^{-\mu_m\tau_m} & 0 \end{pmatrix},$$

$$V = \begin{pmatrix} q_1 + \mu_h & 0 & 0 \\ 0 & \eta_h + q_2 + \mu_h & 0 \\ 0 & 0 & \mu_m \end{pmatrix}.$$

According to [44, Lemma 2.2] and [36, Corollary 2.1], $\mathcal{R}_m$ and $\mathcal{R}_0$ for system (3.11) are as follows:

$$\begin{aligned} \mathcal{R}_m &= \tilde{\rho}(\tilde{F}\tilde{V}^{-1}) = \frac{\mu_b\kappa_d\eta_w e^{-\mu_L\tau_w}}{(\kappa_d+\mu_E)(\eta_w+\mu_E)\mu_m}, \\ \mathcal{R}_0 &= \tilde{\rho}(FV^{-1}) = \sqrt{\frac{\rho\beta_h S_h^* e^{-\mu_h\tau_h}\beta_m S_m^* e^{-\mu_m\tau_m}}{\mu_m(q_1+\mu_h)} + \frac{(1-\rho)\beta_h S_h^* e^{-\mu_h\tau_h}\beta_m S_m^* e^{-\mu_m\tau_m}}{\mu_m(\eta_h+q_2+\mu_h)}}, \end{aligned} \tag{3.12}$$

where $S_h^* = \Lambda_h/\mu_h$, $S_m^* = K_d\Big[(\eta_w\kappa_d e^{-\mu_L\tau_w})/((\eta_w+\mu_E)\mu_m) - (\kappa_d+\mu_E)/\mu_b\Big]$.

4. Numerical simulations

From 2014 to the end of September 2018, Brazil has reported 697,564 CHIKV cases, which is the country with the highest number of CHIKV cases among 51 countries in the Americas [12,45]. The northeast region of Brazil is the area with the greatest risk in the disease [46]. Between January 2016 and December 2017, the state of Ceará, located in northeastern Brazil, reported the largest CHIKV outbreak in its history, with more than 80,000 cases [47,48]. Aedes aegypti, the main vector of the disease, is disseminated in all Brazilian states [46,49], and is greatly affected by climatic factors such as temperature and rainfall. Thus, we focus on the impact of these two climate factors on the spread of the disease in the state of Ceará and the effectiveness of different control strategies.

4.1. Parameter estimates.

Estimation of constant parameters. In view of the Census Organization of Brazil 2010 (see https://www.ibge.gov.br/estatisticas/sociais/populacao/9662-censo-demografico-2010.html?edicao=9665&t=series-historicas), the population of Ceará

Table 2
Constant parameter values in model (2.5).

Parameters	Value	Dimension	Reference
Λ_h	9483.5715	Humans $\times$ Month^{-1}	See text
μ_h	0.1122%	Month^{-1}	See text
N_h^*	8,452,381	Dimensionless	See text
α_h	0.13	Dimensionless	[26]
α_m	0.3	Dimensionless	[26]
τ_h	6/30.4	Month	[50]
ρ	0.2	Dimensionless	[3]
q_1	1/6	Month^{-1}	[51]
q_2	1/9	Month^{-1}	[51]
η_h	[0,1]	Dimensionless	[3,52]
K_d	$2N_h^*$	Dimensionless	[19]
μ_E	0.3344	Month^{-1}	[16,53]

Table 3
Monthly mean temperature and rainfall of Ceará from 1901 to 2016 (in $^\circ$C and *mm*).

Month	January	February	March	April	May	June
Temperature	26.92	26.82	25.81	25.71	24.96	25.19
Rainfall	57.7	100.51	145.16	115.18	58.24	26.10
Month	July	August	September	October	November	December
Temperature	25.35	26.03	27.34	27.53	27.51	28.04
Rainfall	15.77	13.38	9.62	9.20	12.48	24.28

state is 8,452,381, which is used as the value of N_h^*. According to the CIA WorldFact Book (see https://www.cia.gov/library/publications/resources/the-world-factbook/geos/br.html), the life expectancy of Brazil is 74.3 years, and the natural mortality rate of humans is estimated as follows:

$$\mu_h = \frac{1}{74.3 \times 12} = 0.1122\%\ \text{Month}^{-1}.$$

Then the recruitment rate Λ_h for Ceará state can be calculated by

$$\Lambda_h = \mu_h \times N_h^* = 9483.5715\ \text{Humans} \times \text{Month}^{-1}.$$

The estimation of constant parameters is listed in Table 2.

Estimation of temperature-dependent parameters. The monthly average temperature and rainfall of Ceará state used to estimate time-dependent parameters are available on Climate Change Knowledge Portal website (https://climateknowledgeportal.worldbank.org/country/brazil/climate-data-historical), which is listed in Table 3. And we will use the toolbox CFTOOL to fit the time-dependent parameters.

From [31,54,55], the temperature-dependent EIP (in days), daily biting rate and mortality rate per day of mature mosquitos are respectively

$$\begin{aligned}
\tau_m(T) &= 4 + \exp(5.15 - 0.123T)\ \text{Day},\\
\beta(T) &= 9.43 \times 10^{-2} + 4.3 \times 10^{-3}T\ \text{Day}^{-1},\\
\mu_m(T) &= 0.8692 - 0.1590T + 1.116 \times 10^{-2}T^2 - 3.408 \times 10^{-4}T^3 + 3.809 \times 10^{-6}T^4\ \text{Day}^{-1},
\end{aligned}$$

where and thereinafter T is the temperature in $^\circ$C. Thus, the time-dependent $\tau_m(t)$, $\beta(t)$ and $\mu_m(t)$ in Ceará state are fitted, respectively, by

$$\begin{aligned}
\tau_m(t) = {} & 0.3529 - 0.035\cos(\pi t/6) + 0.003868\cos(\pi t/3) - 0.001026\cos(\pi t/2)\\
& - 0.003847\cos(2\pi t/3) - 0.001816\cos(5\pi t/6) + 0.01288\sin(\pi t/6)\\
& + 0.001501\sin(\pi t/3) - 0.002311\sin(\pi t/2) - 0.0008803\sin(2\pi t/3)\\
& + 0.005149\sin(5\pi t/6)\ \text{Month},
\end{aligned}$$

$$\begin{aligned}
\beta(t) = {} & 6.151 - 1.124\cos(\pi t/6) - 0.09214\cos(\pi t/3) - 0.06907\cos(\pi t/2)\\
& + 0.02766\cos(2\pi t/3) - 0.01823\cos(5\pi t/6) - 0.3742\sin(\pi t/6)\\
& - 0.3113\sin(\pi t/3) + 0.04815\sin(\pi t/2) + 0.004922\sin(2\pi t/3)\\
& + 0.004693\sin(5\pi t/6)\ \text{Month}^{-1},\\
\mu_m(t) = {} & 0.8913 - 0.0738\cos(\pi t/6) + 0.008316\cos(\pi t/3) - 0.00082\cos(\pi t/2)\\
& - 0.007358\cos(2\pi t/3) - 0.003511\cos(5\pi t/6) + 0.02745\sin(\pi t/6)
\end{aligned}$$

$$+0.003467\sin(\pi t/3)-0.005849\sin(\pi t/2)-0.001454\sin(2\pi t/3)$$
$$+0.01108\sin(5\pi t/6)\ \text{Month}^{-1}.$$

According to Valdez et al. [16], Wang and Zou [56], the temperature-dependent mature delay (in days), birth rate of dry eggs, maturation rate of wet eggs to larvae and mortality rate of larvae per day are as follows:

$$\tau_w(T) = -236.87+49.38T-3.2515T^2+8.85\times10^{-2}T^3-8.633\times10^{-4}T^4\ \text{Day},$$
$$\mu_b(T) = 0.0591(-5.4+1.8T-0.2124T^2+1.015\times10^{-2}T^3-1.515\times10^{-4}T^4)\ \text{Day}^{-1},$$
$$\eta_w(T) = 8.054\times10^{-4}(T+273.15)\frac{\exp\left[5.4338\times10^3\left(\frac{1}{298}-\frac{1}{T+273.15}\right)\right]}{1+\exp\left[5.0322\times10^4\left(\frac{1}{14184}-\frac{1}{T+273.15}\right)\right]}\ \text{Day}^{-1},$$
$$\mu_L(T) = 2.130-0.3797T+2.457\times10^{-2}T^2-6.778\times10^{-4}T^3+6.794\times10^{-6}T^4\ \text{Day}^{-1}.$$

Then the time-dependent $\tau_w(t)$, $\mu_b(t)$, $\eta_w(t)$ and $\mu_L(t)$ in Ceará state can be fitted by

$$\begin{aligned}\tau_w(t) = {} & 0.3205-0.03366\cos(\pi t/6)+0.005958\cos(\pi t/3)-0.0008135\cos(\pi t/2)\\ & -0.00266\cos(2\pi t/3)-0.0001281\cos(5\pi t/6)+0.01178\sin(\pi t/6)\\ & -0.0001059\sin(\pi t/3)-0.001936\sin(\pi t/2)-0.001072\sin(2\pi t/3)\\ & +0.005026\sin(5\pi t/6)\ \text{Month},\\ \mu_b(t) = {} & 13+1.497\cos(\pi t/6)-0.1712\cos(\pi t/3)+0.03474\cos(\pi t/2)\\ & +0.1574\cos(2\pi t/3)+0.07228\cos(5\pi t/6)-0.5515\sin(\pi t/6)\\ & -0.06285\sin(\pi t/3)+0.1049\sin(\pi t/2)+0.03533\sin(2\pi t/3)\\ & -0.2218\sin(5\pi t/6)\ \text{Month}^{-1},\\ \eta_w(t) = {} & 8.092+0.6638\cos(\pi t/6)-0.04064\cos(\pi t/3)+0.02173\cos(\pi t/2)\\ & +0.08795\cos(2\pi t/3)+0.0581\cos(5\pi t/6)-0.253\sin(\pi t/6)\\ & -0.05107\sin(\pi t/3)+0.04846\sin(\pi t/2)+0.01313\sin(2\pi t/3)\\ & -0.09654\sin(5\pi t/6)\ \text{Month}^{-1},\\ \mu_L(t) = {} & 1.776+0.01859\cos(\pi t/6)-0.0159\cos(\pi t/3)+0.0001412\cos(\pi t/2)\\ & -0.004046\cos(2\pi t/3)-0.008949\cos(5\pi t/6)-0.00294\sin(\pi t/6)\\ & +0.008987\sin(\pi t/3)-0.001066\sin(\pi t/2)+0.002087\sin(2\pi t/3)\\ & -0.00309\sin(5\pi t/6)\ \text{Month}^{-1}.\end{aligned}$$

Note that the change of monthly mean temperature of Ceará is not very large. Combining the above temperature-dependent and mosquito-related mathematical formulas, we then see that the values of the corresponding function do not change much. This also explains that the amplitude of the time-dependent parameter fluctuations fitted by CFTOOL is relatively small.

Estimation of rain-dependent parameter. It follows from [16,57] that the fraction of dry eggs that become wet eggs per day and the number of rainy days every month are given by

$$\kappa_d(R) = 0.8\frac{(R/7.5)^5}{1+(R/7.5)^5}\ \text{Day}^{-1},$$
$$D(M) = 14(0.5+0.5\cos(\pi/6(M-3)))^3+1\ \text{Day},$$

where R is the rainfall in *mm*, M is the month and $M=1,2,3,4,5,6,7,8,9,10,11,12$. Then we obtain the time-dependent fraction of dry eggs $\kappa_d(t)$:

$$\begin{aligned}\kappa_d(t) = {} & 3.034-0.1738\cos(\pi t/6)-1.69\cos(\pi t/3)+0.2186\cos(\pi t/2)\\ & +0.2288\cos(2\pi t/3)-0.1448\cos(5\pi t/6)+3.671\sin(\pi t/6)\\ & -0.1724\sin(\pi t/3)-0.5534\sin(\pi t/2)+0.1659\sin(2\pi t/3)\\ & +0.1544\sin(5\pi t/6)\ \text{Month}^{-1}.\end{aligned}$$

4.2. Model validation

Long term behaviors for system (2.5). We can conduct numerical calculation of the considerable threshold parameters $\mathcal{R}_m$ and $\mathcal{R}_0$ by [36]. The initial values are $S_h(\psi)=446,100$, $A_h(\psi)=7,170$, $I_h(\psi)=46,230$, $E_d(0)=15,870,000$, $E_w(\psi)=17,987,000$, $S_m(\psi)=38,961,000$, $I_m(\psi)=175,000$, $\forall\psi\in[-\tilde{\tau},0]$, then $E_m(0)\approx 33,808$. In Fig. 2, the control measures, such as the use of pesticides and aggressive puddle-clearing, result in the change of the birth rate and hatching fraction of dry eggs and mortality rate of adult mosquitoes into $0.3\mu_b(t)$, $0.4\kappa_d(t)$ and $1.8\mu_m(t)$, respectively. It then follows that

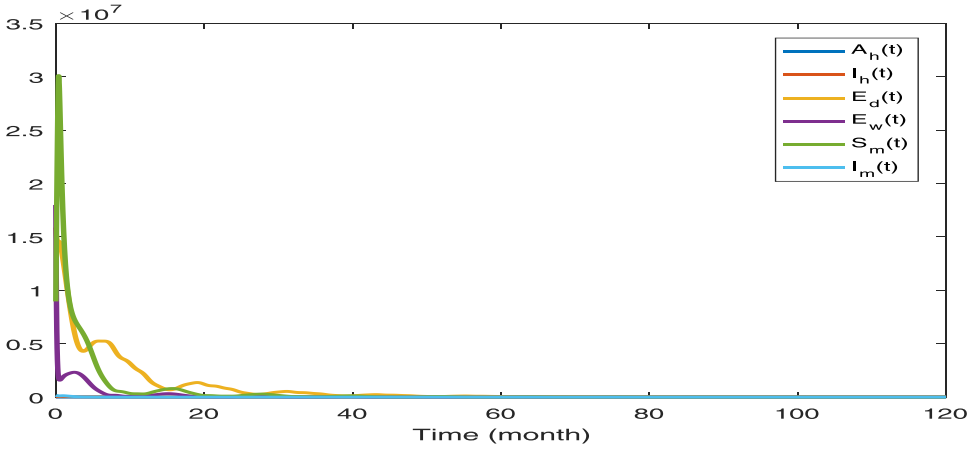


Fig. 2. The time series plot of system (2.5) when $\mathcal{R}_m = 0.9595 < 1$. Here the egg birth rate, hatching fraction of dry egg and mortality rate of female mature mosquitoes are set to be $0.3\mu_b(t)$, $0.4\kappa_d(t)$ and $1.8\mu_m(t)$, respectively.

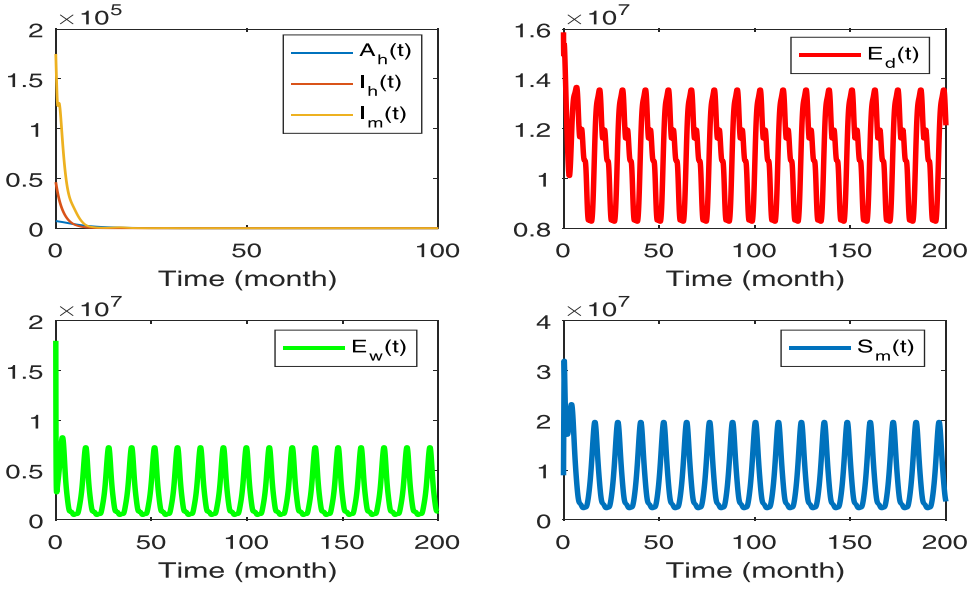


Fig. 3. The time series plots of system (2.5) when $\mathcal{R}_m = 1.4115 > 1$ and $\mathcal{R}_0 = 0.9554 < 1$. Here the egg birth rate, hatching fraction of dry egg, mortality rate of female mature mosquitoes, bite rate and hospitalizing rate are set to be $0.7\mu_b(t)$, $0.7\kappa_d(t)$, $1.5\mu_m(t)$, $0.75\beta(t)$ and $\eta_h = 0.5$, respectively.

$\mathcal{R}_m = 0.9595 < 1$, and hence, Aedes aegypti mosquitoes are eliminated from this area and the disease is not present. In Fig. 3, suppose the effectiveness of the above control measures makes the above three parameters become $0.7\mu_b(t)$, $0.7\kappa_d(t)$ and $1.5\mu_m(t)$, respectively. Moreover, by using of insecticide-treated bed nets and wearing long-sleeved clothing and active medical treatment, result in a bite rate of $0.75\beta(t)$ and a hospitalizing rate η_h of 0.5, then $\mathcal{R}_m = 1.4115 > 1$, $\mathcal{R}_0 = 0.9554 < 1$. Hence, populations of immature and susceptible mosquitoes fluctuate periodically. At the same time, infectious people and mosquitoes tend to 0, and the disease is eliminated from Ceará. If nothing is done to control mosquitoes and humans during the Chikungunya outbreak, then we obtain $\mathcal{R}_m = 1.8583 > 1$, $\mathcal{R}_0 = 5.2020 > 1$. From Fig. 4, one can clearly observe the periodic fluctuations of all the compartments, and the disease will remain endemic in humans and mosquitoes for a long time in Ceará state. Therefore, the results of Theorem 3.1 are well verified by Figs. 2–4.

Effects of rainfall on mosquito population and human infection. If the influence of rainfall on Aedes aegypti is not considered, that is, $\kappa_d = 0$, then egg E is $E_d + E_w$, thus subsystem (3.1) becomes

$$\begin{cases} \frac{dE}{dt}(t) = \mu_b(t)\left(1 - \frac{E(t)}{K_d(t)}\right)S_m(t) - (\eta_w(t) + \mu_E(t))E(t), \\ \frac{dS_m}{dt}(t) = (1 - \tau_w'(t))\eta_w(t - \tau_w(t))E(t - \tau_w(t))e^{-\int_{t-\tau_w(t)}^{t}\mu_L(\xi)d\xi} - \mu_m(t)S_m(t). \end{cases}$$

Compared with Fig. 5(a) and (b), we see that the number of infectious humans and mosquitoes would have been overestimated if rainfall is not taken into account. Besides, in order for the mosquitoes to die out, we assume that the effectiveness

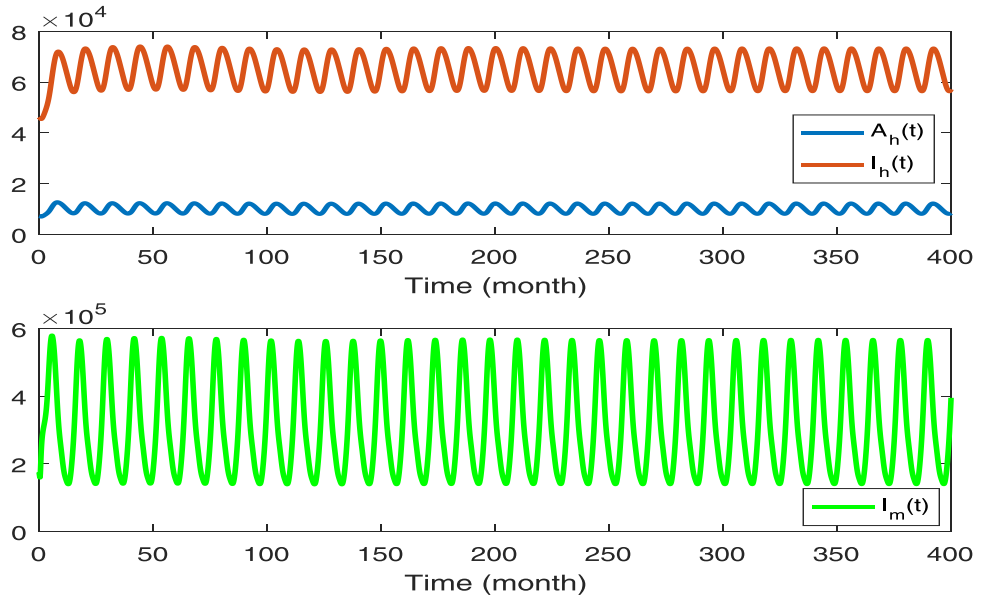


Fig. 4. The time series plots of system (2.5) when $\mathcal{R}_m = 1.8583 > 1$ and $\mathcal{R}_0 = 5.2020 > 1$. Here the egg birth rate, hatching fraction of dry egg, mortality rate of female mature mosquitoes, bite rate and hospitalizing rate are set to be $\mu_b(t)$, $\kappa_d(t)$, $\mu_m(t)$, $\beta(t)$ and $\eta_h = 0$, respectively.

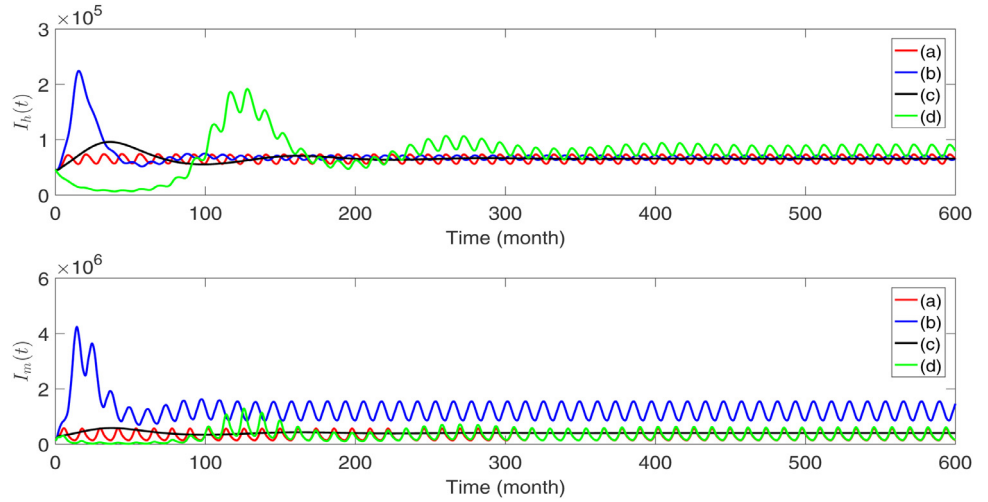


Fig. 5. Comparison of the long-term behavior of infectious people and mosquitoes in different cases: (a) All three factors are taken into account; (b) Rainfall is not considered; (c) Seasonality is not considered; (d) Asymptomatic infection is not considered.

of larvicide is $\mathcal{N}_m$ $(0 \le \mathcal{N}_m \le 1)$, and hence, the birth rate becomes $(1-\mathcal{N}_m)\mu_b(t)$. The birth rate would have to go down to $(1-0.8451)\mu_b(t)$ when the rainfall factor was taken into account, and go down to $(1-0.8743)\mu_b(t)$ when this factor is not taken into account, see Fig. 6(a) and (b). We assume that the effectiveness of insecticide-treated bed net is $\mathcal{N}_0$ $(0 \le \mathcal{N}_0 \le 1)$, and hence, the bite rate becomes $(1-\mathcal{N}_0)\beta(t)$. To eliminate the disease, the bite rate should be kept below $(1-0.8078)\beta(t)$ when rain is taken into account, and below $(1-0.883)\beta(t)$ when this factor is not taken into account, see Fig. 7(a) and (b). This means that without considering the effect of rainfall, we need to invest more resources to reduce mosquitoes and hence the transmission rate of the disease. In addition, if the hatching fraction of dry egg turns to $(1-\mathcal{N}_k)\kappa_d(t)$ by clearing puddles and the hatching fraction is below $(1-0.9794)\kappa_d(t)$, then we have $\mathcal{R}_m < 1$, see Fig. 8(a), which shows that it is very difficult to exterminate mosquitoes through this strategy in tropical rainy areas.

Effects of seasonality on mosquito population and human infection. Following [44], we denote the average of a continuous periodic function $f(t)$ with the period ω by

$$[f] := \frac{1}{\omega}\int_0^\omega f(t)dt.$$

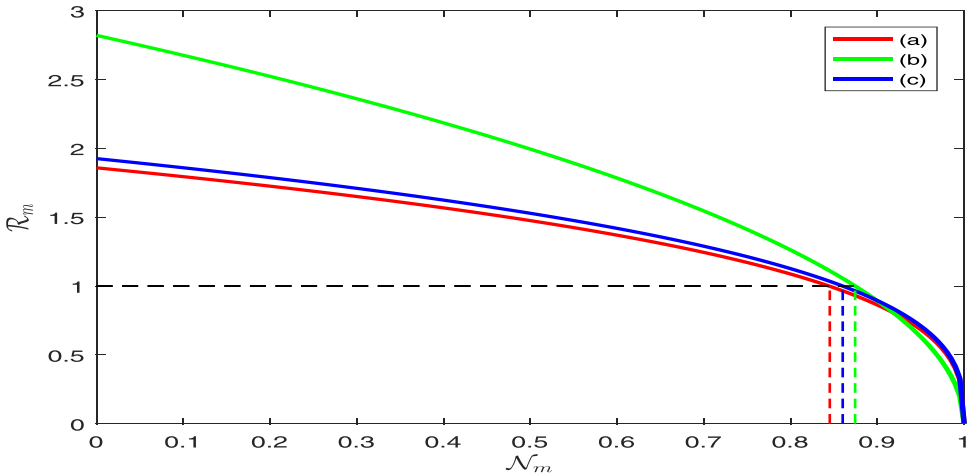


Fig. 6. $\mathcal{R}_m$ varies with $\mathcal{N}_m$ in different cases: (a) Both factors are taken into account; (b) Rainfall is not considered; (c) Seasonality is not considered.

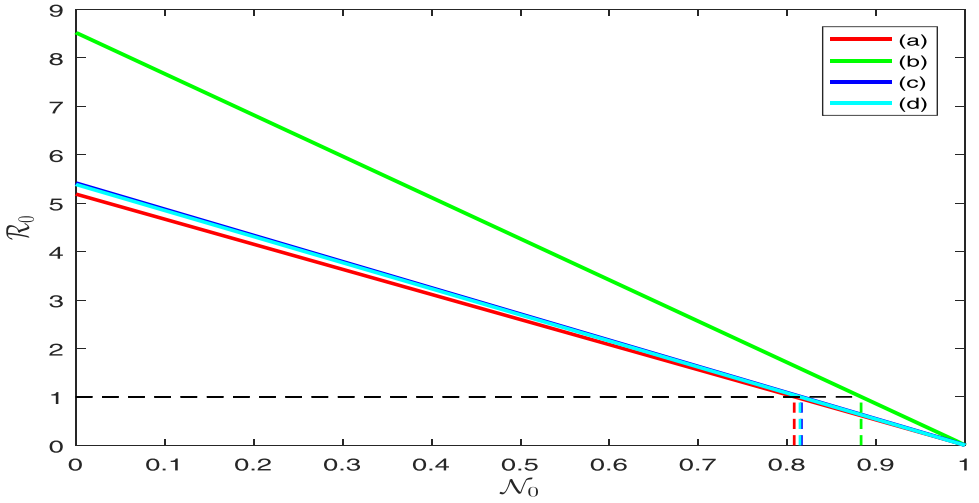


Fig. 7. $\mathcal{R}_0$ varies with $\mathcal{N}_0$ in different cases: (a) All three factors are taken into account; (b) Rainfall is not considered; (c) Seasonality is not considered; (d) Asymptomatic infection is not considered.

Then the time-averaged parameter values of system (2.5) are $[\mu_b] = 13$, $[\kappa_d] = 3.034$, $[\tau_w] = 0.3205$, $[\eta_w] = 8.092$, $[\mu_L] = 1.776$, $[\tau_m] = 0.3529$, $[\beta] = 6.151$ and $[\mu_m] = 0.8913$. We can use the above parameter values and formula (3.12) to calculate $[\mathcal{R}_m] = 1.9252$ and $[\mathcal{R}_0] = 5.4325$ for system (3.11), which is more than 1.8583 and 5.2020 in Fig. 4, respectively. From Fig. 6(a) and (c) and Fig. 7(a) and (c), compared with $\mathcal{R}_m$ and $\mathcal{R}_0$, we see that $[\mathcal{R}_m]$ and $[\mathcal{R}_0]$ overestimates the mosquito population and the infection risk, which can also be found in Fig. 5(a) and (c).

In Fig. 9, we use PRCC (Partial rank correlation coefficient) to obtain the sensitivity analysis of $[\mathcal{R}_m]$ and $[\mathcal{R}_0]$. From Fig. 9(a), we see that $[\mathcal{R}_m]$ is more sensitive to $[\tau_w]$, $[\mu_b]$ and $[\kappa_d]$, followed by $[\mu_L]$ and $[\mu_E]$. Further, Fig. 9(b) indicates that $[\mathcal{R}_0]$ is more sensitive to $[\alpha_h]$, $[\alpha_m]$ and $[\mu_m]$, followed by $[\tau_w]$ and $[\eta_h]$. Thus, the corresponding control measures should be taken for these sensitive parameters.

Effects of asymptomatic infection on CHIKV transmission. If $\rho = 0$, that is, infectious people are not classified as symptomatic or asymptomatic, then system (3.5) becomes

$$\begin{cases} \frac{dI_h}{dt}(t) = -c_{22}(t)I_h(t) + c_{23}(t)I_m(t-\tau_h), \\ \frac{dI_m}{dt}(t) = c_{31}(t)(A_h(t-\tau_m(t)) + I_h(t-\tau_m(t))) - c_{33}(t)I_m(t). \end{cases} \tag{4.1}$$

We can define $\mathcal{R}_0$ for system (4.1) in a similar way. There are two interesting things: on one hand, the basic reproduction ratio $\mathcal{R}_0$ is overestimated from Fig. 5(a) and (d); on the other hand, the number of infective humans is also overestimated from Fig. 7(a) and (d). In addition, $\mathcal{R}_0$ is a decreasing function of ρ, which implies that $\mathcal{R}_0$ is sensitive to ρ (see Fig. 8(b)).

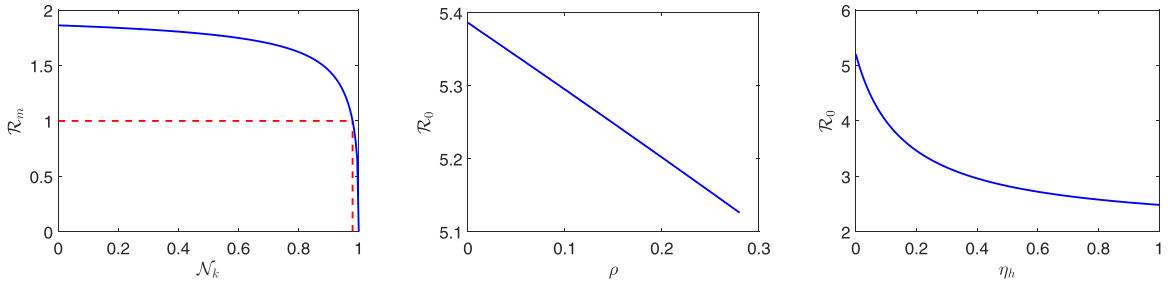


Fig. 8. (a) $\mathcal{R}_m$ as a function of $\mathcal{N}_k$; (b) $\mathcal{R}_0$ as a function of ρ; (c) $\mathcal{R}_0$ as a function of η_h.

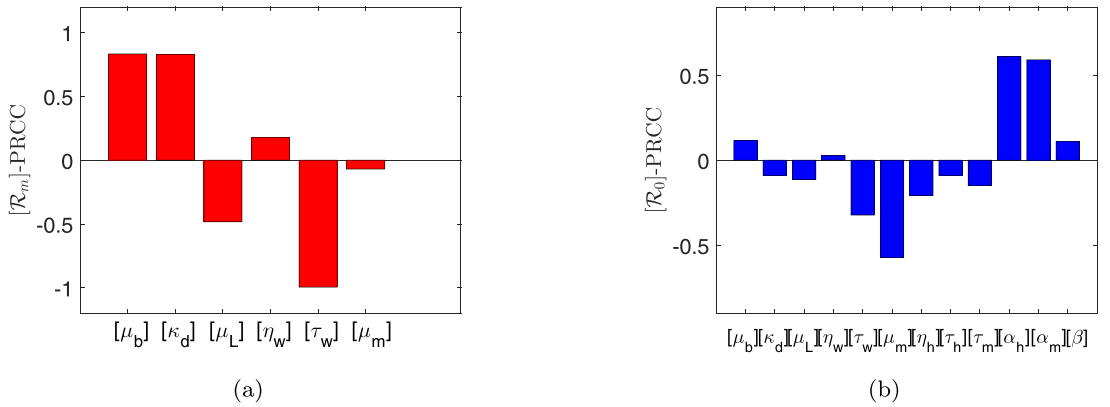


Fig. 9. (a) Sensitivity analysis of $[\mathcal{R}_m]$; (b) Sensitivity analysis of $[\mathcal{R}_0]$.

The effects of hospitalizing rate η_h on $\mathcal{R}_0$. In Fig. 8(c), $\mathcal{R}_0$ decreases as η_h increases and CHIKV is not eliminated when the rate reaches 100 %, which is because the symptoms of the disease are similar to those of dengue and Zika, its misdiagnosis rate is high [52].

5. Discussions

In this paper, we propose a periodic Chikungunya model with temperature and rainfall effects, which takes into account time-dependent EIP, time-dependent maturation delay, asymptomatic and symptomatic infectious humans. According to the theory developed in [36], we derive mosquito reproduction number $\mathcal{R}_m$ and basic reproduction number $\mathcal{R}_0$. Using the theoretical method in [37], we first prove that linear systems (3.2) and (3.5) generate periodic semiflows on the defined phase spaces, respectively, and these semiflows are eventually strongly monotone. Then we show that $\mathcal{R}_m$ and $\mathcal{R}_0$ are threshold parameters for the extinction and persistence of Aedes aegypti and Chikungunya virus, respectively.

In order to verify the theoretical results via numerical simulations, we use the temperature and rainfall data of Ceará state of Brazil, where Chikungunya has been prevalent in recent years. Figs. 2–4 are in perfect agreement with the results of Theorem 3.1. Then we study the effects of rainfall, seasonality, and asymptomatic infection in humans on mosquito population and the CHIKV transmission. By comparing the long-term behavior of infectious variables (see Fig. 5), mosquito reproduction number $\mathcal{R}_m$ (see Fig. 6) and basic reproduction number $\mathcal{R}_0$ (see Fig. 7) in different situations, we find that mosquito population and disease risk are both overestimated without considering these factors. Moreover, Fig. 8(a) and (b) also show that $\mathcal{R}_m$ and $\mathcal{R}_0$ are sensitive to the factors by using these two threshold parameters as functions of $\mathcal{N}_k$ and ρ, respectively. Further, if actively removing puddles is regarded as control strategies, its effect needs to reach more than 0.9794 to eliminate mosquitoes (see Fig. 8(a)). Fig. 8(c) implies that vector control is particularly important in CHIKV prevention in the absence of vaccines, and the integrated Aedes vector control should be a top priority. Besides, we use PRCC to obtain the sensitivity analysis of $[\mathcal{R}_m]$ and $[\mathcal{R}_0]$ (see Fig. 9).

At the moment we have not obtained the monthly new CHIKV cases for the Ceará state, so we are unable to compare the infected numbers predicted by our model with the real data. In addition, we have ignored the duration of the egg phase and the effect of rainfall on the mosquito's other phases (see [16,58]). And we would like to leave these problems for future investigation.

CRediT authorship contribution statement

Xianning Liu: Conceptualization, Methodology, Writing - review & editing. **Yan Wang:** Conceptualization, Methodology, Formal analysis, Software, Writing - original draft. **Xiao-Qiang Zhao:** Conceptualization, Methodology, Writing - review & editing.

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